

Causal Inference on Quantiles in High Dimensions: A Bayesian Approach

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Abstract

This paper proposes a novel Bayesian Analog of Doubly Robust (BADR) procedure for estimating unconditional Quantile Treatment Effects (QTEs) in observational studies. This estimand offers valuable insights into treatment effect heterogeneity across the outcome distribution. By leveraging Bayesian regularization techniques together with efficient-influence-function-based posterior corrections implemented through Bayesian bootstrap, the proposed framework can effectively handle high-dimensional controls and nonlinear relationships while delivering accurate uncertainty quantification. BADR achieves asymptotic equivalence to efficient frequentist estimators as established by a semiparametric Bernstein–von Mises theorem, which permits flexible nuisance estimation under a double-robust smoothness tradeoff between the propensity score and conditional outcome distribution. Monte Carlo simulations show that BADR estimators yield lower RMSE, more accurate coverage probabilities, and shorter credible interval lengths relative to popular alternative estimators found in the literature. We revisit the role of microcredit expansion and loan access on Moroccan household outcomes, demonstrating how the new method adds value in characterizing heterogeneous distributional impacts on outcomes and detecting changes in overall economic inequality, which is also appealing to other applied contexts.

Keywords: Quantile Treatment Effects; Bayesian Inference; Double Robustness; Regularization; BART; Shrinkage Priors.

JEL classifications: C11, C31, D63, G51.

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1 Introduction

When evaluating the causal effect of policy interventions, the distributional impact appeals to researchers and policymakers rather than the average impact alone. It helps to gain more comprehensive and nuanced understanding of the complex effects, ultimately leading to more effective decision-making. In many instances, uniform policies may benefit certain individuals while adversely affecting others. If the effects are considerably heterogeneous, the average treatment effect may not be a sufficient measure, as it likely masks substantial positive and negative effects. Consequently, it is crucial to determine whether certain individuals are worse off as a result of the policy. Even if multiple programs generate positive effects for all individuals, the one that offers the greatest benefits to those at the lower tail of the distribution of the outcome variable might be the most favorable. To illustrate, consider two job training programs with identical mean net impact that is positive. The first program, which increases wages at the bottom of the wage distribution, would be more appreciated than the second program, which only raises the top of the wage distribution. This necessitates the advancement of econometric techniques to enable studies on distributional treatment effects in the presence of heterogeneity. This goal has received special interest and has become increasingly relevant in economic applications¹.

Sets of quantile treatment effects (QTEs) can characterize the heterogeneous impacts of the treatment on different points of the outcome distribution. With a binary treatment, as originally defined by [Doksum \(1974\)](#) and [Lehmann et al. \(1974\)](#), QTEs measure the difference between the unconditional quantiles of the potential outcome distribution under treatment and the potential outcome distribution under nontreatment. Put differently, this captures any difference between the two cumulative distribution functions of treated and untreated potential outcomes. Moreover, the quantile method may be employed, even by those not primarily interested in distributional consequences, to enhance the robustness of their analysis. This is particularly relevant in light of the well-established fact that median regression is more resistant to outliers than mean regression, while many economic data sets involve heavy tails. One notable example in development economics is household welfare measures, including consumption and business outcomes. The difference between these two unconditional distributions of the potential outcomes itself might be appealing to policymakers. This reflects the change in the distribution function as a whole when the treatment could be exogenously shifted between two distinct counterfactual scenarios: universal treatment and no treatment. As the entire distribution function often yields insights into inequality or social welfare analysis, computing QTEs serves as a convenient way to summarise noteworthy aspects. For instance, this enables the detection of changes in overall inequality in the distribution of outcomes, which is a critical concern given the potentially negative consequences of social and economic inequality in the contemporary world².

¹A variety of relevant applications include, but are not limited to, financial interventions ([Meager, 2022](#); [Callaway and Li, 2019](#)), educational subsidies ([Duflo et al., 2021](#)), public health policies ([Schiele and Schmitz, 2016](#)), and local migration incentives ([Chetty et al., 2016](#); [Bryan et al., 2014](#)); all entail social welfare implications and garner substantial attention in public discourse.

²This interest is at the core of the econometric literature strand on distributional counterfactual analysis (see, e.g., [Firpo and Pinto, 2016](#); [Chernozhukov et al., 2013](#); [Rothe, 2010](#)).

In this paper, the primary focus is on unconditional QTEs, which are separate from conditional QTEs. Although both are standard parameters of interest in the program evaluation literature, it is important to highlight the distinction between them. An unconditional (marginal) quantile function is a one-dimensional function of the quantile level τ only. Defined as the difference between the unconditional quantiles of the treated and untreated potential outcome distributions, unconditional QTEs describe the effects of treatment status on the overall outcome distribution without conditioning on the covariates. In contrast, conditional quantile functions are multi-dimensional, depending on not only a chosen quantile level but also values of the covariates. Conditional QTEs thereby express the effects on the outcome distribution within sub-populations characterized by covariate values. More specifically, an individual may rank high in the unconditional distribution of the outcome, meanwhile possessing a low rank in the conditional distribution of the outcome. This is possible if that person has values of observed characteristics that are associated with a large value of outcome overall, yet within the group of people sharing identical values of the observed characteristics, he or she has a comparatively low outcome³. Conditional QTEs enable examination of the heterogeneity of the effects with respect to the observables; however, they might be sensitive to the choice of covariates to be included. Unconditional QTEs, on the other hand, aggregate the conditional effects across the entire population, thereby being more easily conveyed to the policymakers and the public, at the cost of not providing any information about the relationship between the covariates and the outcome. Further discussion can be found in surveys by [Glewwe and Todd \(2022\)](#) and [Frölich and Melly \(2013\)](#). The unconditional quantile treatment effects are appropriate estimands to focus on when the ultimate objective is related to the marginal distribution, for example, the welfare of the (unconditionally) poor. This unconditional effect has been a central estimand of interest in the literature on micro-credit expansion and housing outcomes, reinforcing the need for robust methods suited to recovering and exploring heterogeneity in these effects.

When the target causal estimand is an unconditional quantile treatment effect (QTE), several identification strategies have been developed in the literature. One common approach relies on the assumption of exogenous treatment, typically in the form of a randomized controlled experiment where all participants comply with their treatment assignment. In this ideal scenario, implementing an unconditional QTE estimator is a straightforward process, similar to estimating the average treatment effect (ATE) directly from the realized outcomes of control and treatment groups. However, when such experimental data is unavailable, inferring causal relationships from observational data poses challenges because the observed treatment status is not assigned randomly. This gives rise to the second approach based on the *selection-on-observables* assumption, which implies that the treatment is as good as randomly assigned once we condition on observables. This assumption is widely applicable to empirical studies in economics, such as when the treatment is randomly allocated within demographic groups. It is worth noting that, although our ultimate goal is to obtain an unconditional QTE, covariate information serves to correctly identify the unconditional quantiles and remove selection bias. In our paper, we maintain this identifying assumption of

³Consider a simple example involving wages and years of education, the median income of all individuals with doctoral degrees may be greater than the top quantile for high school dropouts, presuming a strong positive association between education levels and earnings.

selection-on-observables. As elaborated in the subsequent sections, employing covariates for the sake of identification involves identifying the entire conditional cumulative distribution function (CDF) of each potential outcome conditional on potentially high-dimensional covariates. This CDF is then a nuisance function for the identification of QTEs. More often than not, applied researchers encounter a vast set of possible covariates, but they are uncertain about which specific ones are necessary to control for when recovering treatment effects. In addition, the conditional CDF can itself be a complex function. This necessitates the consideration of high-dimensional models to estimate quantile treatment effects.

This paper aims to circumvent such obstacles and contribute to the emerging econometric literature on identification and estimation of QTEs. We propose a novel framework, the Bayesian Analog of Doubly Robust (BADR) approach, for estimating QTEs in an observational study while accounting for the presence of potentially high-dimensional covariates. Briefly, we employ Bayesian techniques to specify and estimate both the treatment assignment and the outcome models, obtaining posterior draws that are then plugged into the doubly robust estimator for QTEs. This estimator is derived as the solution to efficient influence functions, leading to its double-robustness property. The resulting BADR framework comprises two ingredients. First, to effectively accommodate high-dimensional covariates and nonlinear relationships while achieving proper uncertainty quantification, we incorporate various Bayesian regularization methods including sparsity-inducing priors and Bayesian nonparametric methods to generate auxiliary estimators for both the propensity score and the conditional distribution of the outcome variable. We leverage multiple Bayesian quantile regressions to address the unique challenge in quantile estimation, which differs from previous studies on ATE. Second, our method provides double protection against model specification by employing posterior predictive distributions of parameters from both the treatment assignment and outcome models. In the absence of high-dimensional covariates, this approach collapses to a doubly-robust Bayesian estimator for the QTEs without shrinkage priors, which itself has not been explored in previous literature. Overall, the proposed strategy enables us to obtain QTE estimators which showcase highly flexible Bayesian modeling manner coupled with favorable frequentist properties in finite samples.

The advantages of the proposed estimators are demonstrated in Monte Carlo simulations, which consider difficult settings such as high-dimensional covariate spaces or complex nonlinear effects of covariates. Through numerical evidence, we observe substantial gains in bias reduction for QTE estimates across all scenarios, highlighting the strong estimation and inferential features of our methodology in comparison with the naive estimator and popular approaches considered in the literature. The proposed methodology introduces a fresh perspective to empirical research by offering a novel approach for estimating unconditional QTEs in microeconomic applications. The new estimator, Bayesian Doubly Robust with shrinkage prior, allows us to revisit the microcredit experiment originating from the work of Crépon et al. (2015) and explore the impact of household financial access on household welfare. Unlike previous studies that strictly rely on the randomization of microcredit availability at the village level, we employ a new causal estimand and identification strategy that utilizes observed, non-random borrowing patterns at the household level as well as observable household characteristics. Our findings indicate an overall positive effect, with

heterogeneous impacts across the different points of each outcome distribution of interest. It is anticipated that universal financial access will result in an ex-post increase in economic inequality among households, mostly attributed to the significant improvements in consumption and business outcomes at the upper quantiles. Notably, there is evidence of systematic harm in terms of total profit, as a segment of households may experience adverse effects that extend the left tail of the distribution to the left.

Related Literature

Causal Inference on Quantiles

This paper contributes to the emerging literature on unconditional quantile treatment effects (QTEs). [Firpo \(2007\)](#) first considered efficient estimation of unconditional quantile treatment effects (QTEs) and proposed an inverse propensity weighting (IPW) estimator based on propensity scores estimated using a sieve approach, specifically a logistic power series approximation. Under strong smoothness conditions, this IPW estimator is $\sqrt{N}$ -consistent⁴ and achieves the semiparametric efficiency bound, which is reminiscent of analogous results for the IPW estimator for the average treatment effect (ATE) with nonparametrically estimated propensities ([Hirano et al., 2003](#)). Although these purely weighted methods circumvent the estimation of nuisances that depend on the estimand, their desired behavior is restricted to certain nonparametric weight estimators and requires strong smoothness assumptions. Extending the IPW estimator to high-dimensional settings runs into issues due to the fact that its convergence rate can be slowed down by that of the propensity score and its error may depend heavily on the particular method used to learn the propensity score. The properties prohibit the use of general machine learning methods and potentially leading to unstable estimates. In this sense, [Firpo \(2007\)](#)’s IPW estimator lacks the double robustness and flexibility of our proposal.

[Zhang et al. \(2012\)](#) developed several nonparametric methods that resemble those used for ATE and proved that the augmented inverse probability weighted (AIPW) estimator, by augmenting a term that involves the residuals from the outcome regression model, enhances the efficiency of the IPW estimator. The AIPW estimator is expected to be locally efficient and doubly robust under regularity conditions. [Díaz \(2017\)](#) proposed a semiparametric approach using targeted maximum likelihood estimation (TMLE) for marginal quantiles. While sharing the same asymptotic properties as the standard AIPW, the TMLE estimator demonstrates better finite-sample performance when analyzing causal effects on the quantiles, similar to improvements in the mean effect (e.g., [Van der Laan et al., 2011](#)). Our proposed approach relates to both AIPW and TMLE estimators when solving the estimating equation derived from the efficient influence function, a core concept for achieving double robustness. However, [Zhang et al. \(2012\)](#)’s AIPW method assumes strong distributional assumption (e.g., a normal linear model after a Box-Cox transformation of the outcome for each treatment), limiting its application to cases of positive outcomes. In contrast, our Bayesian Analog of Doubly Robust estimation framework employs Bayesian data-adaptive estimation to flexibly fit nuisance functions. [Díaz \(2017\)](#)’s TMLE approach can be considered quite general and closest to our approach among frequentist methods. While both opt for estimating the conditional distribution as

⁴ N is the sample size.

an important middle step, the distinct feature of our modeling strategy lies in utilizing multiple Bayesian quantile regressions and thus can incorporate regularization seamlessly.

Unlike previous studies that did not explicitly consider the case of potentially high-dimensional covariates, [Kallus et al. \(2024\)](#) proposed Localized Debiased Machine Learning (LDML) to enable efficient inference on QTEs in this scenario. For ATE estimation, nuisance functions do not depend on the estimand and can therefore be estimated independently using flexible, data-driven, machine learning methods and plugged into the estimating equation. This Debiased Machine Learning (DML) approach is, however, far more challenging for QTEs estimation, as the efficient influence function involves nuisances that depend on the estimand of interest. Specifically, DML requires we learn the whole conditional cumulative distribution function of a real-valued outcome, potentially conditioned on high-dimensional covariates, evaluated at the quantile of interest. To obviate this cumbersomeness, [Kallus et al. \(2024\)](#) localize the nuisance estimation step to a single initial rough guess of the estimand, such as the IPW estimate, thereby enabling the standard use of machine learning methods in this DML-extended framework. Despite also aiming for a flexible modeling manner, our paper takes a different approach, estimating the whole continuum of the estimand-dependent nuisances by discretizing a hypothetical continuum of quantile regression estimators. The rationale for our choice is based on the advantages of Bayesian quantile regression over the frequentist alternative.

While most studies on unconditional QTEs are based on frequentist methods, [Xu et al. \(2018\)](#) proposed a Bayesian nonparametric approach (BNP) that utilizes a Bayesian additive regression tree (BART) model to estimate the propensity score, followed by a Dirichlet process mixture (DPM) of normals model to construct the distribution of potential outcomes conditional on the estimated propensity score. A key advantage of this approach over frequentist methods is the simultaneous estimation of multiple quantiles of interest. However, it can be regarded as a propensity score analysis which avoids directly modeling the conditional distribution of potential outcomes given the covariates. In contrast, we propose Bayesian Analog of Doubly Robust estimators that can handle a large number of covariates and are more robust to misspecifications.

Causal Inference in High Dimensions

This paper fits into a broader literature on high-dimensional causal inference with observational data. High-dimensional settings are becoming increasingly prevalent, presenting challenges for causal inference. This problem involves either a large number of available covariates or an outcome model with an infinite or large number of parameters, such as nonparametric and semiparametric models. Regularization, a popular technique originally designed to perform prediction in high-dimensional data analysis, has garnered substantial attention in causal inference. It gives rise to numerous causal machine-learning techniques which provide high-quality inference on treatment effects ([Belloni et al., 2014](#); [Chernozhukov et al., 2018](#); [Farrell, 2015](#); [Athey et al., 2018](#)). While the majority of studies are frequentist regularization-based approaches, there has been growing interest in adopting Bayesian regularization-based techniques into causal inference, as Bayesian inference is a natural probabilistic framework for quantifying uncertainty and learning about model parameters. It is known that many frequentist penalized likelihood estimators can be considered equivalent to the posterior modes of

Bayesian estimators under certain choices of shrinkage priors such as spike-and-slab prior (Mitchell and Beauchamp, 1988; Ishwaran and Rao, 2005), Bayesian Lasso prior (Park and Casella, 2008), and Horseshoe prior (Carvalho et al., 2009, 2010).⁵ Recent studies have successfully deployed these techniques for confounding adjustment to estimate average treatment effects in the presence of high-dimensional controls (Antonelli et al., 2019, 2022; Hahn et al., 2018). Bayesian nonparametric methods are also powerful tools utilized for regularization within the Bayesian paradigm. Among them, the Bayesian Additive Regression Tree (BART) has emerged as a workhorse widely used for causal inference. Introduced by Chipman, George, and McCulloch (2010; 2006), BART models offer several advantages over linear models, such as automatic adaptation to nonlinearity. Regarding implementation, BART is also preferred due to its fast computation, good performance of default choices of hyperparameters and available software (Linero and Antonelli, 2023). When there is sufficient covariate overlap, BART has been shown to outperform numerous Frequentist machine learning methods in prediction problems, including random forests. Hill (2011) further proposed the use of BART in causal inference and demonstrated its efficacy in flexibly modeling the response surface. To mitigate the regularization-induced confounding issue (Hahn et al., 2018) when using a BART outcome model, Hahn et al. (2020) developed the Bayesian causal forest model, a BART-based approach that includes a fixed estimate of the propensity score for additional adjustment in the outcome model. This model yields excellent performance in estimating heterogeneous treatment effects, making BART a strong default choice for integrating Bayesian nonparametric methods into causal effect modeling. Subsequent studies by Spertus and Normand (2018) and Xu et al. (2018) employed BART models to fit the propensity score in the first stage, enabling the use of Bayesian propensity score analysis to estimate ATE (with high-dimensional data) and QTEs, respectively.

Double Robustness

This paper also adds to the development of doubly robust estimators, which have gained extensive use in the causal inference literature (Scharfstein et al., 1999; Bang and Robins, 2005) owing to their desirable property of providing consistent inference even under misspecification of either the treatment assignment or outcome regression models (but not both). For a comprehensive survey of doubly robust estimators and their properties, we refer interested readers to Daniel (2014).

Doubly robust estimators have been extended to accommodate nonparametric or high-dimensional settings by enabling data-adaptive estimation of treatment and outcome models. This includes doubly robust estimators with the group Lasso (Farrell, 2015), double machine learning estimators (Chernozhukov et al., 2018), doubly robust matching estimators (Antonelli et al., 2018) and targeted maximum likelihood estimators (Van der Laan et al., 2011), among others. In these complex settings, doubly robust estimators offer an extra benefit that parametric convergence rates ($\sqrt{N}$) can be achieved even when each of the propensity score or outcome regression models converges at slower rates ($\sqrt[4]{N}$ or faster). Roughly speaking, this echoes the insights from recent advances in the double machine learning literature (Chernozhukov et al., 2022). Specifically, penalizing either the

⁵Thorough reviews and well-designed simulations could be found in Korobilis and Shimizu (2022), Van Erp et al. (2019) and Polson and Sokolov (2019), who advocate the merits of Bayesian sparsity-inducing priors in comparison to frequentist counterparts.

propensity score model or the outcome model alone would be insufficient for valid causal inference, but combining the two as nuisance functions achieves a desirable convergence rate and finite-sample performance in high-dimensional causal analyses.

There have been attempts to propose doubly robust Bayesian recipes, however, this area is still less established than its Frequentist counterpart. This is mainly due to a lack of consensus on propensity score adjustment in the Bayesian causal modeling framework (Robins and Ritov, 1997; Robins et al., 2015; Li et al., 2023), despite its central role being well recognized in the literature (Rosenbaum and Rubin, 1983; Zigler, 2016). Pioneering work on the Bayesian approach for doubly robust causal inference was done by Saarela et al. (2016), in which the authors formalized the problem and addressed it by combining the posterior predictive distribution of parameters with the Bayesian bootstrap. The idea of utilizing posterior predictive distribution was later advanced in the line of work by Antonelli et al. (2022) and Shin and Antonelli (2023), who aimed to improve inference for doubly robust estimators for the average treatment effect (ATE) and the conditional average treatment effect (CATE), respectively. The general strategy involves estimating both the propensity score and the conditional outcome mean using Bayesian methods. Posterior draws from their respective posterior predictive distributions are then obtained and plugged into a doubly robust estimator. While this approach is not fully Bayesian because there is no joint likelihood for all parameters stated, it effectively integrates Bayesian modeling techniques and Frequentist inferential procedures for causal analysis with proper uncertainty quantification. These features are particularly important in high-dimensional scenarios, where handling large numbers of covariates and quantifying uncertainty can be challenging. Recent theoretical developments established Bernstein-von Mises (BvM) results when examining the asymptotic behaviors of Bayesian inference procedures for causal effects. Ray and van der Vaart (2020) obtained BvM for ATE using propensity score-adjusted priors, but that prior adjustment alone leaves a bias when smoothness is traded off between the propensity score and the conditional outcome mean functions. Breunig et al. (2025) make this explicit and introduce a posterior correction to address the bias term and prove a semiparametric BvM under double-robust smoothness conditions. Relatedly, Yiu et al. (2025) demonstrated that using double robust estimands still requires posterior corrections for valid inference, even when their conditions for the BvM theorem impose Donsker properties for the propensity score and the conditional mean function – a more restrictive assumption than the prior correction of Ray and van der Vaart (2020) and particularly, the double robust method of Breunig et al. (2025). A comparison between the double robust method and semiparametric Bayesian methods was also provided in Breunig et al. (2024) for average treatment effects on the treated (ATT) in Difference-in-Differences (DiD) context. Both satisfy the BvM, but only the regularized version achieves it under double robust smoothness conditions. In more regular, low-dimensional setting, the semiparametric Bayesian approach is very competitive, but its performance declines with increasing model complexity. This motivates double robust approaches coupled with posterior corrections in complex designs.

In spite of the progress made in Bayesian literature, most studies have focused on the doubly robust estimation of either unconditional or conditional average treatment effect. Our paper presents a distinctive contribution by concentrating on unconditional quantile treatment effects (QTE). This causal estimand is of independent interest as it offers a different and complementary

approach to uncover treatment effect heterogeneity. Although the conditional average treatment effect (CATE) also characterizes treatment effect heterogeneity, the effects vary across individuals or subgroups defined by observed characteristics. In contrast, unconditional QTE focuses on the effect heterogeneity of the treatment across different outcome ranks without conditioning on individual characteristics or covariates. Evaluating the impact on the entire outcome distribution of interest makes this approach particularly relevant to distributional concerns and inequality, offering valuable insights when used alongside other estimands, such as CATE, to understand the potential consequences of treatments and policies.

Paper Organization

The remainder of this paper is organized as follows. In Section 2, we formally define quantile treatment effects in a causal framework along with key identification assumptions and present the proposed estimation and inference approach. Section 3 establishes main theoretical results. Next, we evaluate the performance of our methods using simulations in Section 4 and use the proposed method to examine the causal impact of loan access on the distribution of household outcomes in Section 5. Finally, we conclude the paper in Section 6 with brief final remarks on the method and policy recommendations based on our results.

2 Setup and Implementation

2.1 Observed Data and Causal Estimand

Let $Z_i = (Y_i, T_i, X_i)$, $i = 1, \dots, n$, be independent and identically distributed (i.i.d.) observations generated from an unknown probability law $P_0 \in \mathcal{P}$. Here $Y_i \in \mathbb{R}$ is the observed outcome, $T_i \in \{0, 1\}$ is a binary treatment indicator, and $X_i \in \mathcal{X} \subseteq \mathbb{R}^p$ is a p -dimensional vector of covariates. For a measurable function h , we adopt the standard empirical process notations

$$P_0 h := E_0\{h(Z)\} = \int h(z) dP_0(z), \quad \mathbb{P}_n h := \frac{1}{n} \sum_{i=1}^n h(Z_i), \quad \mathbb{G}_n h = \sqrt{n}(\mathbb{P}_n - P_0)h.$$

For $t \in \{0, 1\}$, let $Y(t)$ denote the potential outcome under treatment t . Define the marginal counterfactual distribution and its τ -quantile by

$$F_{t0}(y) := P_0\{Y(t) \leq y\}, \quad (2.1)$$

$$q_{t0}(\tau) := F_{t0}^{-1}(\tau) = \inf\{y : F_{t0}(y) \geq \tau\}, \quad \tau \in (0, 1). \quad (2.2)$$

The causal parameter of interest is the marginal quantile treatment effect

$$\Delta_{\tau,0} := q_{10}(\tau) - q_{00}(\tau) = F_{10}^{-1}(\tau) - F_{00}^{-1}(\tau). \quad (2.3)$$

Thus, $\Delta_{\tau,0}$ compares the same quantile index across the two marginal potential-outcome distributions. Without additional restrictions on the joint distribution of $\{Y(1), Y(0)\}$, it is not the τ -quantile of the individual treatment effect $Y(1) - Y(0)$.

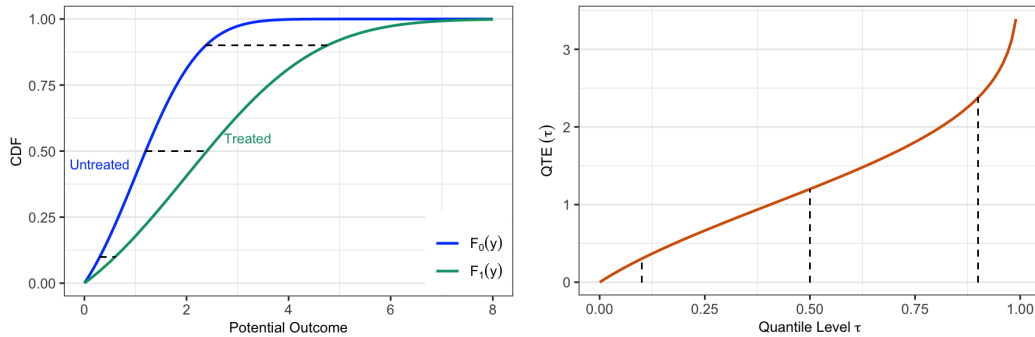


Figure 2.1: Illustration of Quantile Treatment Effects (QTEs). The left-hand figure demonstrates unconditional distributions of Treated and Untreated potential outcomes, which are colored in green and blue, respectively. The horizontal distance between these two distributions yields QTEs. For instance, $\text{QTE}(0.95)$, $\text{QTE}(0.5)$, and $\text{QTE}(0.05)$ are represented by three dashed lines in the figure. QTEs across all values of quantile levels are plotted in the right-hand figure.

For the remainder of the theoretical development, τ is fixed and we write $q_{t0} = q_{t0}(\tau)$ when no ambiguity arises.

2.2 Identification and Efficient Influence Function

For any probability law $P \in \mathcal{P}$, we define the arm-specific propensity score and conditional observed-outcome distribution by

$$\pi_{t,P}(x) := P(T = t \mid X = x), \quad (2.4)$$

$$G_{t,P}(y \mid x) := P(Y \leq y \mid T = t, X = x). \quad (2.5)$$

Under the true law P_0 , we write

$$\pi_{t0}(x) = \pi_{t,P_0}(x), \quad G_{t0}(y \mid x) = G_{t,P_0}(y \mid x).$$

We impose the standard identification conditions below.

Assumption 2.1 (Identification). *The following conditions hold:*

(i) **Unconfoundedness (Selection-on-Observables).**

$$Y^{(1)}, Y^{(0)} \perp\!\!\!\perp T \mid X, \quad (2.6)$$

where $\perp\!\!\!\perp$ denotes statistical independence. This stipulates that the treatment is independent of the potential outcome (i.e. as good as randomly assigned) conditioned on observables X .

(ii) **Strong Overlap.** *There exists $c_\pi \in (0, 1/2)$ such that*

$$c_\pi \leq \pi_{10}(X) = P_0(T = 1 \mid X) \leq 1 - c_\pi \quad (2.7)$$

This states that all subjects have a positive probability of receiving any treatment level. This is a necessary condition to estimate treatment effects everywhere in covariate space.

(iii) **Stable Unit Treatment Value Assumption (SUTVA).** *The observed outcome satisfies*

$$T_i = t \text{ implies } Y_i = Y_i^{(t)}, \quad \text{for } t \in \{0, 1\}. \quad (2.8)$$

This ensures that for each subject i , the same treatment level cannot lead to different observed outcomes [Little and Rubin \(2000\)](#). We observe the potential outcome corresponding to the realized treatment level. Also, one unit's treatment does not affect another unit's potential outcomes.

Under Assumption 2.1, the counterfactual CDF is identified by

$$\begin{aligned} F_{t0}(y) &= E_0[P_0\{Y(t) \leq y \mid X\}] \\ &= E_0[P_0\{Y(t) \leq y \mid T = t, X\}] \\ &= E_0[P_0(Y \leq y \mid T = t, X)] = E_0\{G_{t0}(y \mid X)\}. \end{aligned} \quad (2.9)$$

The second equality follows from *Unconfoundedness* and the third from *SUTVA*. *Overlap* ensures

that $G_{t0}(y | x)$ is identified over covariate values represented in the target population.

For candidate nuisance functions $\eta_t = (\pi_t, G_t)$ with π_t bounded away from zero, define the augmented distributional score

$$\phi_t(Z; y, \eta_t) := G_t(y | X) + \frac{\mathbb{1}\{T = t\}}{\pi_t(X)} \{ \mathbb{1}\{Y \leq y\} - G_t(y | X) \}. \quad (2.10)$$

Lemma 2.1 (Exact doubly robust identity). *For every y ,*

$$P_0 \phi_t(\cdot; y, \eta_t) - F_{t0}(y) = E_0 \left[\frac{\pi_t(X) - \pi_{t0}(X)}{\pi_t(X)} \{ G_t(y | X) - G_{t0}(y | X) \} \right]. \quad (2.11)$$

Hence, the right-hand side is zero if either $\pi_t = \pi_{t0}$ or $G_t(y | \cdot) = G_{t0}(y | \cdot)$.

Proof. Conditioning on X gives

$$E_0 \{ \phi_t(Z; y, \eta_t) | X \} = G_t(y | X) + \frac{\pi_{t0}(X)}{\pi_t(X)} \{ G_{t0}(y | X) - G_t(y | X) \}.$$

Subtracting $G_{t0}(y | X)$ and then taking expectations proves (2.11). $\square$

Suppose F_{t0} is differentiable at q_{t0} with marginal density $f_{t0}(q_{t0}) > 0$. The efficient influence function (EIF) for the marginal quantile $q_{t0}(\tau)$ in the nonparametric observed-data model is

$$\varphi_{t0,\tau}(Z) = - \frac{\phi_t(Z; q_{t0}, \eta_{t0}) - \tau}{f_{t0}(q_{t0})}. \quad (2.12)$$

Equivalently,

$$\varphi_{t0,\tau}(Z) = - \frac{1}{f_{t0}(q_{t0})} \left[\frac{\mathbb{1}\{T = t\}}{\pi_{t0}(X)} \{ \mathbb{1}\{Y \leq q_{t0}\} - G_{t0}(q_{t0} | X) \} + G_{t0}(q_{t0} | X) - \tau \right].$$

In particular, $E_0 \{ \varphi_{t0}(Z) \} = 0$ and $V_{t0} = E_0 \{ \varphi_{t0}^2(Z) \}$ is the arm-specific semiparametric efficiency bound. The QTE efficient influence function and efficiency bound are

$$\varphi_{\Delta,0}(Z) = \varphi_{10}(Z) - \varphi_{00}(Z), \quad V_{\Delta,0} = E_0 \{ [\varphi_{10}(Z) - \varphi_{00}(Z)]^2 \}. \quad (2.13)$$

2.3 BADR-QTE Point Estimators and Credible Sets

With the primary parameter of interest being the quantile treatment effect (QTE), we develop a Bayesian analog of doubly robust (BADR) inference framework for this causal estimand. The construction builds on the general one-step posterior correction of [Yiu et al. \(2025\)](#). The central idea is to replace the initial estimator in classical one-step estimation by a draw from an initial posterior and to replace the empirical measure in the correction term by an independent draw from the Bayesian bootstrap posterior. This produces an entire corrected posterior distribution, rather than only a bias-corrected point estimator. The counterfactual quantile problem requires additional

ingredients. The target is obtained by inverting an estimated counterfactual distribution, and its influence function depends on an arm-specific propensity score, a conditional outcome distribution evaluated at a random plug-in quantile, and the marginal density at that quantile. Our framework thus combines the target-specific objects with a high-dimensional nuisance posterior, finite-grid conditional-CDF reconstruction, and a joint two-arm correction that preserves cross-arm covariance.

Let

$$Z^{(n)} = (Z_1, \dots, Z_n), \quad Z_i = (Y_i, T_i, X_i) \stackrel{\text{i.i.d.}}{\sim} P_0.$$

Let $\Pi(\cdot \mid Z^{(n)})$ denote an initial posterior on $\mathcal{P}$. A draw from this posterior is denoted by

$$P^{(b)} \sim \Pi(\cdot \mid Z^{(n)}).$$

Let $\Pi_{\text{BB}}(\cdot \mid Z^{(n)})$ denote the Bayesian bootstrap posterior, and write

$$\tilde{P}^{(b)} = \sum_{i=1}^n W_i^{(b)} \delta_{Z_i}, \quad (W_1^{(b)}, \dots, W_n^{(b)}) \sim \text{Dirichlet}(1, \dots, 1).$$

Thus, $\tilde{P}^{(b)}$ is a random probability measure supported on the observed sample.

Following [Yiu et al. \(2025\)](#), consider a real-valued, pathwise differentiable functional $\psi : \mathcal{P} \rightarrow \mathbb{R}$, and let $\dot{\psi}_P$ denote its efficient influence function at P . The generic one-step corrected parameter is

$$\tilde{\psi}(P, \tilde{P}) = \psi(P) + \tilde{P}[\dot{\psi}_P], \quad (2.14)$$

where, conditionally on the observed data,

$$(P, \tilde{P}) \mid Z^{(n)} \sim \Pi(\cdot \mid Z^{(n)}) \times \Pi_{\text{BB}}(\cdot \mid Z^{(n)}). \quad (2.15)$$

Accordingly, a corrected posterior draw can be written as

$$\tilde{\psi}^{(b)} = \psi(P^{(b)}) + \sum_{i=1}^n W_i^{(b)} \dot{\psi}_{P^{(b)}}(Z_i).$$

The initial posterior propagates uncertainty in the plug-in functional $\psi(P)$. Conditionally on $P^{(b)}$, the Bayesian bootstrap captures uncertainty in the first-order correction $P_0[\dot{\psi}_{P^{(b)}}]$ by replacing the unknown distribution P_0 with the random measure $\tilde{P}^{(b)}$. The conditional independence imposed in Equation (2.15) separates these two sources of uncertainty and allows the one-step correction to be applied without modifying the sampler used to generate the initial posterior draws.

We now construct the objects required to specialize this correction to the marginal QTE. Fix $\tau \in (0, 1)$. For posterior draw $P^{(b)}$, and for $t \in \{0, 1\}$, write

$$\pi_{t, P^{(b)}}(x) = P^{(b)}(T = t \mid X = x), \quad G_{t, P^{(b)}}(y \mid x) = P^{(b)}(Y \leq y \mid T = t, X = x).$$

Define

$$F_{t,n}^{(b)}(y) := \mathbb{P}_n G_{t,P^{(b)}}(y | X) = \frac{1}{n} \sum_{i=1}^n G_{t,P^{(b)}}(y | X_i), \quad (2.16)$$

and the posterior plug-in quantile

$$q_{t,\tau,n}^{(b)} = \inf\{y \in \mathbb{R} : F_{t,n}^{(b)}(y) \geq \tau\}. \quad (2.17)$$

Let $d_{t,\tau,n}^{(b)} > 0$ denote the draw-specific estimate of the marginal density at $q_{t,\tau,n}^{(b)}$. Its finite-difference construction is given in Subsection 2.4.2.

For $Z = (Y, T, X)$, define

$$\phi_{t,\tau,n}^{(b)}(Z) := \frac{\mathbb{1}\{T = t\}}{\pi_{t,P^{(b)}}(X)} \left[\mathbb{1}\{Y \leq q_{t,\tau,n}^{(b)}\} - G_{t,P^{(b)}}(q_{t,\tau,n}^{(b)} | X) \right] + G_{t,P^{(b)}}(q_{t,\tau,n}^{(b)} | X), \quad (2.18)$$

$$\varphi_{t,\tau,n}^{(b)}(Z) := - \frac{\phi_{t,\tau,n}^{(b)}(Z) - \tau}{d_{t,\tau,n}^{(b)}}. \quad (2.19)$$

Thus the plug-in quantile, conditional CDF, propensity score, and influence function denominator are evaluated at the same nuisance draw.

Independently of $P^{(b)}$, draw

$$(W_1^{(b)}, \dots, W_n^{(b)}) \sim \text{Dirichlet}(1, \dots, 1), \quad \sum_{i=1}^n W_i^{(b)} = 1, \quad (2.20)$$

and obtain the corresponding corrected arm-specific quantile and QTE draws as

$$\tilde{q}_{t,\tau}^{(b)} = q_{t,\tau,n}^{(b)} + \sum_{i=1}^n W_i^{(b)} \varphi_{t,\tau,n}^{(b)}(Z_i), \quad t \in \{0, 1\}, \quad (2.21)$$

$$\tilde{\Delta}_\tau^{(b)} = \tilde{q}_{1,\tau}^{(b)} - \tilde{q}_{0,\tau}^{(b)} = q_{1,\tau,n}^{(b)} - q_{0,\tau,n}^{(b)} + \sum_{i=1}^n W_i^{(b)} \left\{ \varphi_{1,\tau,n}^{(b)}(Z_i) - \varphi_{0,\tau,n}^{(b)}(Z_i) \right\}. \quad (2.22)$$

The same Dirichlet weights are used for both treatment arms. This preserves the cross-arm covariance induced by estimating both marginal quantiles from the same observations. Let $\tilde{\Delta}_\tau$ denote a generic corrected QTE draw generated jointly from the modular nuisance posterior and the independent Bayesian-bootstrap distribution. The induced BADR-QTE posterior is

$$\Pi_{n,\Delta,\tau}^{\text{os}} := \mathcal{L}(\tilde{\Delta}_\tau | Z^{(n)}). \quad (2.23)$$

Provided its first moment exists, we use the posterior mean as the BADR point estimator

$$\hat{\Delta}_{n,\tau}^{\text{BADR}} := E(\tilde{\Delta}_\tau | Z^{(n)}). \quad (2.24)$$

The equal-tailed $(1 - \alpha)$ credible interval is

$$C_{n,\tau}^{\text{BADR}}(1 - \alpha) := \left[Q_{\alpha/2}(\Pi_{n,\Delta,\tau}^{\text{os}}), Q_{1-\alpha/2}(\Pi_{n,\Delta,\tau}^{\text{os}}) \right], \quad (2.25)$$

where $Q_\gamma(\Pi)$ denotes the γ -quantile of a probability law Π .

Algorithm 1: One-step BADR posterior sampling scheme for QTEs

Input: number of posterior draws B and quantile index τ .

1. Obtain draws $P^{(1)}, \dots, P^{(B)}$ from the initial posterior, or equivalently obtain the nuisance draws required to evaluate (2.16) and (2.19).
2. For $b = 1, \dots, B$:
 - (a) Draw $(W_1^{(b)}, \dots, W_n^{(b)})$ from $\text{Dirichlet}(1, \dots, 1)$ independently of $P^{(b)}$.
 - (b) For $t = 0, 1$, compute $F_{t,n}^{(b)}$, $q_{t,\tau,n}^{(b)}$, $d_{t,\tau,n}^{(b)}$, and $\{\varphi_{t,\tau,n}^{(b)}(Z_i)\}_{i=1}^n$.
 - (c) Compute $\tilde{\Delta}_\tau^{(b)}$ from (2.22), using the same weights for both arms.

Output: $\{\tilde{\Delta}_\tau^{(1)}, \dots, \tilde{\Delta}_\tau^{(B)}\}$ and the posterior summaries in (2.24)–(2.25).

2.4 Modeling Approach in High Dimensions

At this juncture, we present the high-dimensional modeling framework we use to provide scalable estimation algorithms. We employ Bayesian techniques to specify and estimate the treatment assignment and outcome models separately. To address the challenges posed by high-dimensional feature spaces, we integrate various Bayesian regularization methods into our proposed framework to yield estimators of the nuisance functions corresponding to the propensity score and the conditional distribution of the outcome, denoted as $\hat{\pi}(X)$ and $\hat{G}(y | t, X)$, respectively. For the treatment assignment model, we adopt Bayesian Additive Regression Trees (BART) priors, whose merits have been increasingly recognized (see, e.g., Chipman et al., 2010; Hill, 2011; Hahn et al., 2020; Linero and Antonelli, 2023). For the outcome model, we leverage multiple Bayesian quantile regressions combined with shrinkage priors to explicitly estimate the conditional distribution while accommodating potentially high-dimensional covariates. This combined strategy allows us to develop a flexible and doubly-robust Bayesian estimator with desirable finite-sample frequentist properties. To the best of our knowledge, this modeling framework has not been previously pursued in the literature, resulting in a novel approach to estimation of heterogeneous treatment effects.

2.4.1 Treatment Assignment Model

We fit a binary Bayesian Additive Regression Trees (BART) model using the observations $\{T_i, X_i\}_{i=1}^n$ to model the regression of the treatment assignment on control variables, that is,

$$\pi(x) = P(T = 1 | X = x) = H[f_{\text{BART}}(x)], \quad (2.26)$$

where the link function H is either the standard normal distribution CDF for probit BART or the logistic CDF for the logit BART, and

$$f_{\text{BART}}(x) = \sum_{m=1}^M f_{\text{tree}}(x; \Gamma_m, \mu_m) \text{ are sum of } M \text{ Bayesian regression trees.}$$

For $m \in \{1, \dots, M\}$, Γ_m is a tree structure that consists of a set of splitting rules and a set of terminal nodes; and $\mu = (\mu_{m,1}, \dots, \mu_{m,b_m})$ is a vector of parameters associated with b_m terminal nodes of Γ_m , such that $f_{\text{tree}}(x; \Gamma_m, \mu_m) = \mu_{m,l}$ if x is corresponding to the l^{th} terminal node of Γ_m .

The modeling choices used to implement the BART specification are presented in Appendix B. Once a sequence of B posterior draws for the underlying BART parameters has been obtained, B posterior samples of the arm-specific propensity can be calculated by⁶

$$\begin{aligned} \pi^{(b)}(X_i) &= H \left[\sum_{m=1}^M f_{\text{tree}}(X_i; \Gamma_m^{(b)}, \mu_m^{(b)}) \right], \\ \pi_t^{(b)}(X_i) &= t\pi^{(b)}(X_i) + (1-t)(1 - \pi^{(b)}(X_i)), \quad \text{for } t = 0, 1; \quad b = 1, \dots, B; \quad i = 1, \dots, N. \end{aligned} \quad (2.27)$$

2.4.2 Outcome Model

In contrast to the conditional expectation estimation, directly specifying a high-dimensional model for the entire conditional density can be restrictive and computationally demanding. We instead recover the conditional distribution from a collection of fitted conditional quantiles. This permits us to combine Bayesian quantile regression with shrinkage priors while retaining a flexible representation of distributional heterogeneity. For each $t \in \{0, 1\}$, recall

$$G_t(y | x) = P(Y \leq y | T = t, X = x)$$

and let

$$\mathcal{Q}_t(u | x) := G_t^{-1}(u | x), \quad u \in (0, 1),$$

denote the corresponding conditional quantile function. Accordingly, the conditional CDF admits the representation

$$G_t(y | x) = \int_0^1 \mathbb{1}\{\mathcal{Q}_t(u | x) \leq y\} du. \quad (2.28)$$

That means, the conditional probability of observing an outcome below y equals the proportion of conditional quantiles that do not exceed y . Equation (2.28) allows us to estimate G_t without specifying a separate likelihood for the entire conditional density.

Specifically, for each treatment arm, we fit Bayesian quantile regressions on a grid

$$0 < u_{1,n} < \dots < u_{S_n,n} < 1.$$

⁶For numerical stability, the implementation clips each draw to $[c_{\text{clip}}, 1 - c_{\text{clip}}]$, with $c_{\text{clip}} = 0.01$ in the simulations. This clipping is asymptotically inactive if the true propensity is uniformly interior to that interval and the fitted propensity contracts uniformly.

Let $Z_{t,n} = \{(Y_i, X_i) : T_i = t\}$ denote the arm-specific sample, and let $\tilde{x}$ include an intercept. At quantile index $u_{s,n}$, a posterior coefficient draw $\beta_{t,s}^{(b)}$ produces the fitted conditional quantile

$$\mathcal{Q}_{t,s}^{\text{raw},(b)}(x) = \tilde{x}^\top \beta_{t,s}^{(b)}, \quad s = 1, \dots, S_n. \quad (2.29)$$

Here, shrinkage priors on $\beta_{t,s}$ can be readily applied to regularize the dependence of the conditional quantiles on the covariates. They therefore allow the outcome model to accommodate potentially high-dimensional X , including settings in which the number of candidate controls is large relative to the sample size. This feature is thoroughly reviewed by [Korobilis and Shimizu \(2022\)](#). Details of the shrinkage priors and the Bayesian quantile-regression sampler are provided in Appendices [C](#) and [D](#), respectively.

Because the regressions are fitted separately across quantile levels, their joint working distribution is naturally represented as the product quasi-posterior

$$\hat{\Pi}_{t,n}^Q = \bigotimes_{s=1}^{S_n} \Pi_{t,s}(d\beta_{t,s} \mid Z_{t,n}). \quad (2.30)$$

A draw of the conditional quantile process is assembled by drawing one coefficient vector from every factor in (2.30). This construction preserves the computational advantages of separate quantile regressions: each model can be estimated in parallel, and different shrinkage specifications can be used without altering the subsequent BADR correction.

Separate quantile regressions may produce crossing fitted quantiles. For every b, t , and x , rearrange the raw fitted values to obtain

$$\mathcal{Q}_{t,1}^{M,(b)}(x) \leq \dots \leq \mathcal{Q}_{t,S_n}^{M,(b)}(x). \quad (2.31)$$

The corresponding conditional-CDF draw is reconstructed as

$$G_{t,S_n}^{(b)}(y \mid x) = \frac{1}{S_n} \sum_{s=1}^{S_n} \mathbb{1} \left\{ \mathcal{Q}_{t,s}^{M,(b)}(x) \leq y \right\}. \quad (2.32)$$

Equation (2.32) is the finite-grid analog of (2.28). It defines a valid conditional CDF for every posterior process draw and does not require the reconstructed CDF to be differentiable. The equal-weight form corresponds directly to the regular quantile grid used in the implementation.⁷

In the reported simulations, we use a fixed 99-point grid ($u_s = 0.01, 0.02, \dots, 0.99$), which provides an accurate and computationally manageable finite-sample approximation. The asymptotic theory

⁷For an equally weighted grid, sorting the fitted quantiles does not change the numerical value of (2.32), because the indicator average is invariant to permutations. Rearrangement nevertheless provides a coherent nondecreasing quantile curve and facilitates subsequent interpolation and inversion. Under standard conditions, monotone rearrangement also does not increase the relevant approximation error relative to a monotone true quantile function; see [Chernozhukov et al. \(2010\)](#).

instead treats the grid as sample-size dependent and requires its discrepancy

$$D_{u,n} = \sup_{u \in [0,1]} \left| \frac{1}{S_n} \sum_{s=1}^{S_n} \mathbb{1}\{u_{s,n} \leq u\} - u \right| \rightarrow 0$$

For a regular interior grid, this follows from $S_n \rightarrow \infty$, $u_{1,n} \rightarrow 0$, $u_{S_n,n} \rightarrow 1$.

For every posterior draw, the reconstructed conditional CDF yields the marginal counterfactual CDF

$$F_{t,n}^{(b)}(y) = \frac{1}{n} \sum_{i=1}^n G_{t,S_n}^{(b)}(y \mid X_i), \quad (2.33)$$

after applying the stated interpolation convention. Inverting (2.33) gives the posterior plug-in quantile $q_{t,\tau,n}^{(b)}$ used as the starting value of the BADR correction.

Finally, rather than differentiating a high-dimensional conditional quantile surface, estimate the marginal density denominator from the one-dimensional marginal CDF

$$d_{t,\tau,n}^{(b)} = \max \left\{ c_d, \frac{F_{t,n}^{(b)}(q_{t,\tau,n}^{(b)} + h_{f,n}) - F_{t,n}^{(b)}(q_{t,\tau,n}^{(b)} - h_{f,n})}{2h_{f,n}} \right\}, \quad (2.34)$$

where $c_d > 0$ prevents numerical division by values arbitrarily close to zero. A one-sided difference is used when an evaluation point reaches the boundary of the outcome grid. The stabilized bandwidth used in the reported simulations is a finite-sample tuning rule. For the asymptotic analysis, it is embedded in a shrinking sequence satisfying the density-estimation rate restrictions imposed in Section 3.

The resulting draws $\{G_{t,S_n}^{(b)}(y \mid x) : b = 1, \dots, B, t = 0, 1\}$ provide the outcome-model component of the initial BADR working posterior. Combined with the corresponding propensity-score draws in (2.27), they determine the posterior plug-in quantiles, draw-specific efficient influence functions, and one-step BADR-QTE posterior described in subsection 2.3. This construction therefore converts independently fitted high-dimensional quantile regressions into the posterior conditional-distribution objects required for doubly robust inference on marginal QTEs.

3 Main Theoretical Results

This section establishes a semiparametric Bernstein–von Mises (BvM) theorem which establishes the asymptotic equivalence between our Bayesian procedure and the frequentist-type semiparametric efficient one. We first consider a marginal potential-outcome quantile $q_t(\tau)$, for $t \in \{0, 1\}$, at a given $\tau \in (0, 1)$, and then establish the joint result required for the quantile treatment effect

$$\Delta_\tau = q_{10}(\tau) - q_{00}(\tau).$$

The result is stated for the target-level one-step posterior obtained by combining draws from the initial modular nuisance posterior with independent Bayesian-bootstrap weights in the efficient-influence-function correction. Its frequentist validity serves as a theoretical justification for our proposed Bayesian procedure. We present the main theoretical results, beginning with a generic BvM theorem formulated under a set of high-level conditions. The result applies broadly and is not restricted to specific prior choices.

Let $\theta = (\pi, G_0, G_1) \in \Theta = \Theta_\pi \times \Theta_{G,0} \times \Theta_{G,1}$ denote the nuisance parameter, where

$$\pi_1(x) = \pi(x), \quad \pi_0(x) = 1 - \pi(x),$$

and

$$G_t(y \mid x) = P(Y \leq y \mid T = t, X = x).$$

Write $\theta_0 = (\pi_0, G_{00}, G_{10})$ for the true nuisance parameter. The modular nuisance posterior is denoted by $\hat{\Pi}_n^\theta(\cdot \mid Z^{(n)})$. The full conditional law used to construct a one-step BADR draw is

$$(\theta, \tilde{P}) \mid Z^{(n)} \sim \hat{\Pi}_n^\theta(\cdot \mid Z^{(n)}) \otimes \Pi_{\text{BB}}(\cdot \mid Z^{(n)}), \quad (3.1)$$

where the two components are conditionally independent. We write $\mathcal{L}_n^{\text{os}}(\cdot \mid Z^{(n)})$ for the conditional law induced by (3.1).

For two positive sequence $\{a_n\}$ and $\{b_n\}$, we write $a_n \lesssim b_n$ if there exists a constant $C > 0$ such that $a_n \leq Cb_n$ for all n . We write $a_n \asymp b_n$ if $a_n \lesssim b_n$ and $b_n \lesssim a_n$. The asymptotic symbol $o_{P_0}(r_n)$ and $O_{P_0}(r_n)$ are defined with respect to the same underlying probability measure P_0 .

3.1 Assumptions

Let $\mathcal{H}_n^\theta \subseteq \Theta$ be a sequence of deterministic measurable sets. Unless stated otherwise, all suprema below are taken over $\theta \in \mathcal{H}_n^\theta$.

Assumption 3.1 (Posterior concentration on regular nuisance sets). *There exists a sequence of measurable sets $\mathcal{H}_n^\theta \subseteq \Theta$ such that*

$$\hat{\Pi}_n^\theta(\mathcal{H}_n^\theta \mid Z^{(n)}) \longrightarrow 1 \quad \text{in } P_0\text{-probability.}$$

Equivalently, for every $\eta > 0$,

$$P_0 \left\{ \hat{\Pi}_n^\theta \left((\mathcal{H}_n^\theta)^c \mid Z^{(n)} \right) > \eta \right\} \longrightarrow 0.$$

Assumption 3.1 states that the modular nuisance posterior asymptotically assigns negligible mass to draws that fail the regularity and rate conditions imposed below. In particular, if

$$\theta^{(b)} \mid Z^{(n)} \sim \hat{\Pi}_n^\theta(\cdot \mid Z^{(n)}),$$

then

$$\Pr \left(\theta^{(b)} \in \mathcal{H}_n^\theta \mid Z^{(n)} \right) = \hat{\Pi}_n^\theta \left(\mathcal{H}_n^\theta \mid Z^{(n)} \right) \longrightarrow 1$$

in P_0 -probability.

Assumption 3.2 (Posterior-draw positivity and local quantile regularity). *Assumption 2.1 holds, so that the true propensity score satisfies the strong-overlap condition in (2.7).*

The target quantile q_{t0} is unique. There exists a neighborhood $\mathcal{N}_t$ of q_{t0} and constants

$$0 < \underline{f} < \bar{f} < \infty$$

such that F_{t0} has three bounded derivatives on $\mathcal{N}_t$ and

$$\underline{f} \leq \inf_{y \in \mathcal{N}_t} f_{t0}(y) \leq \sup_{y \in \mathcal{N}_t} f_{t0}(y) \leq \bar{f}.$$

There exist a constant $c_\star > 0$, deterministic positive sequences $a_n \rightarrow 0$ and $b_n \rightarrow 0$, constants $0 < \underline{d} < \bar{d} < \infty$, and events $\mathcal{E}_n$ satisfying

$$P_0(\mathcal{E}_n) \longrightarrow 1,$$

such that, on $\mathcal{E}_n$, for all sufficiently large n ,

$$\inf_{\theta \in \mathcal{H}_n^\theta} \min_{t \in \{0,1\}} \pi_t(X; \theta) \geq c_\star \quad P_0\text{-almost surely}, \quad (3.2)$$

$$\underline{d} \leq \inf_{\theta \in \mathcal{H}_n^\theta} d_{t,\theta,n} \leq \sup_{\theta \in \mathcal{H}_n^\theta} d_{t,\theta,n} \leq \bar{d}, \quad (3.3)$$

$$\sup_{\theta \in \mathcal{H}_n^\theta} |q_{t,\theta,n} - q_{t0}| \leq a_n, \quad (3.4)$$

$$\sup_{\theta \in \mathcal{H}_n^\theta} |d_{t,\theta,n} - f_{t0}(q_{t0})| \leq b_n. \quad (3.5)$$

Remark 3.1. The strong-overlap condition in Assumption 2.1 concerns the true data-generating distribution and is required for identification. Condition (3.2) instead restricts the candidate propensity-score draws receiving asymptotically nonnegligible posterior mass. This additional condition controls the inverse-propensity factors appearing in the candidate influence functions. In

the implemented procedure, it is enforced by clipping each propensity draw to $[c_{\text{clip}}, 1 - c_{\text{clip}}]$.

No differentiability of the candidate conditional or marginal CDF is required. In particular, $G_t(\cdot | x)$ may be the step function obtained from the finite-grid conditional-quantile reconstruction. Smoothness is imposed on the true marginal distribution F_{t0} , whereas $d_{t,\theta,n}$ is the numerical density denominator constructed from the candidate marginal CDF.

Assumption 3.3 (Nuisance contraction and second-order rates). *There exist deterministic positive sequences $\rho_n \rightarrow 0$ and $\epsilon_n \rightarrow 0$ such that, for all sufficiently large n ,*

$$\sup_{\theta \in \mathcal{H}_n^\theta} \|\pi_t - \pi_{t0}\|_{L^2(P_{0,X})} \leq \rho_n$$

and

$$\sup_{\theta \in \mathcal{H}_n^\theta} \sup_{y \in \mathcal{N}_t} \|G_t(y | \cdot) - G_{t0}(y | \cdot)\|_{L^2(P_{0,X})} \leq \epsilon_n.$$

The rate sequences satisfy

$$\sqrt{n} \left\{ a_n^2 + a_n b_n + \rho_n \epsilon_n \right\} \rightarrow 0. \quad (3.6)$$

Remark 3.2. The term $\rho_n \epsilon_n$ in (3.6) is the doubly robust product remainder. The additional terms a_n^2 and $a_n b_n$ arise from the nonlinear quantile inversion and estimation of the density denominator. In particular, $b_n = o(1)$ alone is insufficient: root- n validity requires

$$\sqrt{n} a_n b_n \rightarrow 0.$$

The primitive rates associated with the implemented BQR reconstruction can be connected to Assumption 3.3 as follows. Suppose the conditional-quantile approximation, quantile-grid discrepancy, and interpolation error satisfy

$$\epsilon_n \lesssim r_{Q,n} + D_{u,n} + \gamma_n, \quad (3.7)$$

where $r_{Q,n}$ controls the conditional-quantile estimation error, $D_{u,n}$ is the discrepancy of the quantile-index grid, and γ_n is the relevant interpolation or outcome-grid mesh. Suppose also that the marginal-CDF error is bounded by $r_{F,n}$, and that the symmetric finite-difference construction satisfies

$$b_n \lesssim h_{f,n}^2 + \frac{r_{F,n}}{h_{f,n}} + a_n, \quad (3.8)$$

for a bandwidth satisfying

$$h_{f,n} \rightarrow 0, \quad \frac{r_{F,n}}{h_{f,n}} \rightarrow 0.$$

Then (3.6) follows from

$$\sqrt{n} \left[a_n^2 + a_n \left\{ h_{f,n}^2 + \frac{r_{F,n}}{h_{f,n}} \right\} + \rho_n \{ r_{Q,n} + D_{u,n} + \gamma_n \} \right] \rightarrow 0. \quad (3.9)$$

Assumption 3.4 (EIF, empirical-process, and bootstrap stability). *Uniformly over $\theta \in \mathcal{H}_n^\theta$,*

$$\|\varphi_{t,\theta,n} - \varphi_{t,\tau,0}\|_{L^2(P_0)} = o(1), \quad (3.10)$$

$$|\mathbb{G}_n(\varphi_{t,\theta,n} - \varphi_{t,\tau,0})| = o_{P_0}(1), \quad (3.11)$$

$$\left| \mathbb{P}_n \left[\{\varphi_{t,\theta,n} - \mathbb{P}_n \varphi_{t,\theta,n}\}^2 \right] - V_{t,\tau} \right| = o_{P_0}(1). \quad (3.12)$$

In addition, the following uniform Lindeberg condition holds:

$$\sup_{\theta \in \mathcal{H}_n^\theta} \frac{\max_{1 \leq i \leq n} |\varphi_{t,\theta,n}(Z_i) - \mathbb{P}_n \varphi_{t,\theta,n}|}{\sqrt{n}} = o_{P_0}(1). \quad (3.13)$$

Condition (3.13), together with (3.12), ensures the conditional central limit theorem for the Bayesian-bootstrap fluctuation

$$\sqrt{n}(\tilde{P} - \mathbb{P}_n)\varphi_{t,\theta,n}.$$

Proposition 3.1 (Uniform second-order remainder). *Suppose Assumptions 3.2 and 3.3 hold. Define*

$$r_{2,t}(P_0, \theta) = q_{t0} - q_{t,\theta,n} - P_0 \varphi_{t,\theta,n}. \quad (3.14)$$

Then, on the events $\mathcal{E}_n$, for all sufficiently large n ,

$$\sup_{\theta \in \mathcal{H}_n^\theta} |r_{2,t}(P_0, \theta)| \leq C \left\{ a_n^2 + a_n b_n + \rho_n \epsilon_n \right\}, \quad (3.15)$$

where $C < \infty$ is independent of n and $\theta \in \mathcal{H}_n^\theta$. Consequently,

$$\sup_{\theta \in \mathcal{H}_n^\theta} \sqrt{n} |r_{2,t}(P_0, \theta)| = o_{P_0}(1). \quad (3.16)$$

3.2 Semiparametric Bernstein–von Mises Theorem

For probability measures P and Q on a metric space $(\mathcal{X}, d)$, define

$$d_{\text{BL}}(P, Q) = \sup_{f \in \text{BL}(1)} \left| \int f d(P - Q) \right|,$$

where

$$\text{BL}(1) = \left\{ f : \sup_x |f(x)| + \sup_{x \neq x'} \frac{|f(x) - f(x')|}{d(x, x')} \leq 1 \right\}.$$

Convergence in d_{BL} metrizes weak convergence.

Let $\hat{q}_{t,\tau}^{\text{eff}}$ be a regular, asymptotically efficient estimator satisfying

$$\sqrt{n} \left(\hat{q}_{t,\tau}^{\text{eff}} - q_{t0} \right) = \mathbb{G}_n \varphi_{t,\tau,0} + o_{P_0}(1). \quad (3.17)$$

Theorem 3.1 (Pointwise BvM theorem for the BADR quantile posterior). *Suppose Assumptions 3.1–3.4 hold. Then*

$$d_{\text{BL}} \left(\mathcal{L}_n^{\text{os}} \left[\sqrt{n} \left(\tilde{q}_{t,\tau} - \hat{q}_{t,\tau}^{\text{eff}} \right) \middle| Z^{(n)} \right], \mathcal{N}(0, V_{t,\tau}) \right) \rightarrow_{P_0} 0. \quad (3.18)$$

The central expansion underlying the theorem is

$$\begin{aligned} \sqrt{n} \left(\tilde{q}_{t,\tau} - \hat{q}_{t,\tau}^{\text{eff}} \right) &= \sqrt{n} (\tilde{P} - \mathbb{P}_n) \varphi_{t,\theta,n} \\ &\quad + \mathbb{G}_n(\varphi_{t,\theta,n} - \varphi_{t,\tau,0}) - \sqrt{n} r_{2,t}(P_0, \theta) + o_{P_0}(1). \end{aligned} \quad (3.19)$$

The first term in (3.19) satisfies the conditional Bayesian-bootstrap central limit theorem. The second and third terms are uniformly negligible by Assumption 3.4 and Proposition 3.1. Assumption 3.1 ensures that these uniform conclusions apply to asymptotically all nuisance posterior draws.

For the joint two-arm result, define

$$\varphi_{\tau,0} = \begin{pmatrix} \varphi_{1,\tau,0} \\ \varphi_{0,\tau,0} \end{pmatrix}, \quad \Sigma_\tau = P_0 \left(\varphi_{\tau,0} \varphi_{\tau,0}^\top \right).$$

Corollary 3.1 (Joint BvM theorem and the BADR–QTE posterior). *Suppose the assumptions of Theorem 3.1 hold jointly for $t = 0, 1$, with the scalar variance and Lindeberg conditions replaced by their corresponding vector versions. Then*

$$d_{\text{BL}} \left(\mathcal{L}_n^{\text{os}} \left[\sqrt{n} \left\{ \begin{pmatrix} \tilde{q}_{1,\tau} \\ \tilde{q}_{0,\tau} \end{pmatrix} - \begin{pmatrix} \hat{q}_{1,\tau}^{\text{eff}} \\ \hat{q}_{0,\tau}^{\text{eff}} \end{pmatrix} \right\} \middle| Z^{(n)} \right], \mathcal{N}_2(0, \Sigma_\tau) \right) \rightarrow_{P_0} 0. \quad (3.20)$$

Let

$$\tilde{\Delta}_\tau = \tilde{q}_{1,\tau} - \tilde{q}_{0,\tau}, \quad \hat{\Delta}_\tau^{\text{eff}} = \hat{q}_{1,\tau}^{\text{eff}} - \hat{q}_{0,\tau}^{\text{eff}}.$$

Then

$$d_{\text{BL}} \left(\mathcal{L}_n^{\text{os}} \left[\sqrt{n} \left(\tilde{\Delta}_\tau - \hat{\Delta}_\tau^{\text{eff}} \right) \middle| Z^{(n)} \right], \mathcal{N}(0, V_{\Delta,\tau}) \right) \rightarrow_{P_0} 0, \quad (3.21)$$

where

$$\varphi_{\Delta,\tau,0} = \varphi_{1,\tau,0} - \varphi_{0,\tau,0}, \quad (3.22)$$

$$V_{\Delta,\tau} = P_0 \varphi_{\Delta,\tau,0}^2 = V_{1,\tau} + V_{0,\tau} - 2P_0(\varphi_{1,\tau,0} \varphi_{0,\tau,0}). \quad (3.23)$$

The use of the same Bayesian-bootstrap weights for both treatment arms is essential for reproducing the covariance term

$$P_0(\varphi_{1,\tau,0} \varphi_{0,\tau,0}).$$

Using independent weights across arms would generally yield the wrong asymptotic variance for the QTE.

Corollary 3.2 (Credible intervals and posterior means). *Suppose the conditions of Corollary 3.1 hold. Then the equal-tailed credible interval $C_{n,\tau}^{\text{BADR}}(1 - \alpha)$ satisfies*

$$P_0 \left\{ \Delta_{\tau,0} \in C_{n,\tau}^{\text{BADR}}(1 - \alpha) \right\} \longrightarrow 1 - \alpha. \quad (3.24)$$

If, in addition, the centered and scaled QTE posterior laws are uniformly integrable—for example, if for some $\delta > 0$,

$$E_n^{\text{os}} \left[\left| \sqrt{n} \left(\tilde{\Delta}_\tau - \hat{\Delta}_\tau^{\text{eff}} \right) \right|^{1+\delta} \middle| Z^{(n)} \right] = O_{P_0}(1), \quad (3.25)$$

then

$$\sqrt{n} \left(\hat{\Delta}_{n,\tau}^{\text{BADR}} - \hat{\Delta}_\tau^{\text{eff}} \right) = o_{P_0}(1),$$

and hence

$$\sqrt{n} \left(\hat{\Delta}_{n,\tau}^{\text{BADR}} - \Delta_{\tau,0} \right) \rightsquigarrow \mathcal{N}(0, V_{\Delta,\tau}).$$

Corollary 3.2 provides a frequentist justification for Bayesian inference based on the BADR–QTE posterior. In particular, posterior means are asymptotically efficient estimators, and Bayesian credible intervals are asymptotically valid frequentist confidence intervals.

4 Simulation Study

We investigate the finite-sample performance of the proposed Bayesian Analog of Doubly Robust (BADR) procedure in an observational setting with high-dimensional confounding. The simulation is designed to assess two distinct aspects of performance. First, we examine the accuracy of point estimation as the dimensionality of the covariate vector becomes large relative to the sample size. Second, we evaluate uncertainty quantification through the joint behavior of empirical coverage probabilities and interval lengths.

We develop two versions of the estimators which represent our proposed methodology – Bayesian Doubly Robust estimator (BDR) and an extension that adds shrinkage priors (BDRS). Specifically, the former employs the original Bayesian Quantile Regression while the latter incorporates the Adaptive Lasso in order to account for sparsity and uncertainty in the outcome model. Both estimators fit the propensity score using a logit BART model in the first step. Furthermore, we also compare our proposed method with existing estimators. The Bayesian nonparametric counterpart (BNP) is a fully Bayesian approach developed in [Xu et al. \(2018\)](#), where the propensity score is estimated using a logit BART, then the conditional distribution of the potential outcome given a BART posterior sample of the propensity score in each treatment group is estimated separately using a Dirichlet process mixture of multivariate normals. We additionally compare three frequentist methods – the Localized Debiased Machine Learning (LDML) method introduced in [Kallus et al. \(2024\)](#), the Targeted Maximum Likelihood Estimation (TMLE) method proposed in [Díaz \(2017\)](#), and Firpo’s Inverse Probability Weighted (FIPW) method developed in [Firpo \(2007\)](#). Among them, LDML and TMLE are two estimators that can leverage a variety of machine learning methods. Particularly, in our simulation exercise, Random Forest is incorporated into LDML and Lasso is integrated into TMLE. Implementation details of these methods can be found in [Appendix E.1](#). The Naïve estimator, which is simply the difference between empirical quantiles of treated and control groups without any adjustment for covariates, serves as a benchmark to illustrate the consequences of ignoring confounding.

In each simulation design, we generate 100 synthetic datasets. For each simulated dataset, quantile treatment effects are estimated at 5 quantile levels

$$\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$$

For Bayesian methods, point estimates are posterior means and uncertainty is summarized by central 95% credible intervals; for frequentist methods, we report the corresponding point estimates and 95% confidence intervals. Point-estimation performance is evaluated using bias, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). For interval estimation, we report empirical coverage probabilities together with average 95% interval lengths.

4.1 Simulation Design

We consider a setting in which the mean of potential outcomes is a linear combination of covariates. We draw 40-dimensional covariates $\mathbf{X}$ ($p = 40$) from the independent standard normal distributions

and allow different sample sizes $N \in \{100, 500, 1000\}$ of the dataset. Consequently, p/N ratio ranges from 0.40 to 0.04, allowing us to evaluate the performance over increasing dimensional difficulty. Treatment assignment depends on a subset of the covariates, while the potential outcomes depend on an overlapping but distinct subset, thereby generating non-random treatment assignment under selection on observables. The exact form of the true model used to generate synthetic data is as follows

$$\begin{aligned}
T \mid \mathbf{X} &\sim \text{Bern}(\pi(\mathbf{X})), \\
Y^{(0)} \mid \mathbf{X} &\sim \mathcal{N}(\mu(\mathbf{X}), 10), \\
Y^{(1)} \mid \mathbf{X} &\sim \mathcal{N}(1 + \mu(\mathbf{X}), 17.5), \\
Y &= T \times Y^{(1)} + (1 - T) \times Y^{(0)}; \\
\text{where } \pi(\mathbf{X}) &= \{1 + \exp[-(X_1 + X_2 + X_3)]\}^{-1}, \\
\mu(\mathbf{X}) &= X_1 + X_2 + X_4 + X_5.
\end{aligned} \tag{4.1}$$

Under this specification, the unconditional distribution of potential outcomes are $Y^{(0)} \sim \mathcal{N}(0, 14)$ and $Y^{(1)} \sim \mathcal{N}(1, 21.5)$. Figure 4.1 provides a visual illustration of the corresponding marginal densities and marginal distributions. As a result, the population quantile treatment effects can be computed analytically. In particular, the true 10th, 25th, 50th, 75th and 90th QTEs are $\Delta_{0.10} = -0.311$, $\Delta_{0.25} = 0.311$, $\Delta_{0.5} = 1$, $\Delta_{0.75} = 1.689$, and $\Delta_{0.90} = 2.311$, respectively.

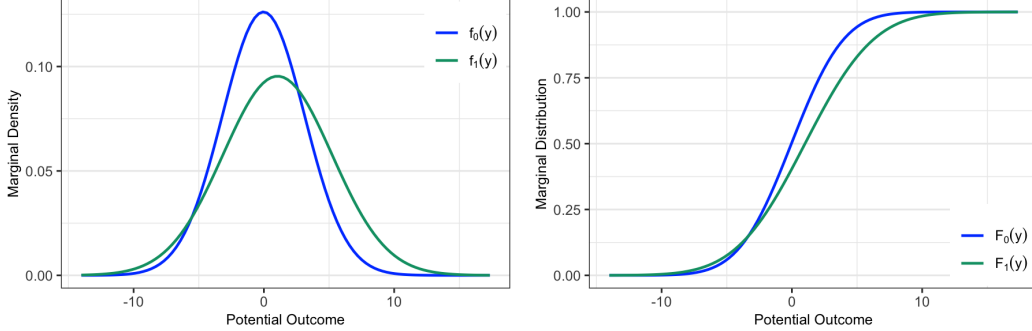


Figure 4.1: True marginal densities and marginal distributions of the treated and untreated potential outcomes in SD1. This design emulates a thought experiment relevant to policy evaluation literature. Hypothetically assigning the entire population either to treatment or to control induces a change in both location and shape of the outcome distribution.

4.2 Simulation Results

Table 4.1 reports the mean QTE estimates for each method across 100 Monte Carlo replications. These computations are based on a sample size of $N = 1000$ and $p = 40$. It is clear that the **Naive** method exhibits substantial and systematic bias throughout the outcome distribution. This can be attributed to the absence of adjustment for confounders in $\mathbf{X}$, resulting in poor performance as expected. In comparison, all other methods considered in our current setting outperform the **Naive** method, proving their effectiveness in correcting selection bias to some extent.

Table 4.1: Mean QTE estimates across 100 Monte Carlo replications ($N = 1000$, $p = 40$).

Method	Percentile				
	10th	25th	50th	75th	90th
True QTE	-0.31	0.31	1.00	1.69	2.31
BDR	-0.36 (2.84)	0.31 (1.62)	0.94 (1.14)	1.65 (1.76)	2.33 (3.41)
BDRS	-0.34 (1.32)	0.31 (1.01)	0.93 (0.85)	1.65 (1.05)	2.32 (1.39)
BNP	0.92 (1.00)	1.55 (0.82)	2.24 (0.75)	2.93 (0.80)	3.59 (1.00)
LDML	0.29 (1.31)	0.96 (1.04)	1.64 (0.85)	2.35 (0.96)	3.05 (1.34)
TMLE	-0.43 (1.57)	0.27 (1.22)	0.92 (1.01)	1.65 (1.15)	2.34 (1.88)
FIPW	-0.38 (2.04)	0.27 (1.87)	0.92 (2.09)	1.64 (2.07)	2.25 (2.61)
Naive	0.94 (1.24)	1.58 (0.97)	2.25 (0.91)	2.97 (0.95)	3.65 (1.21)

Notes: Entries report mean QTE estimates across 100 Monte Carlo replications. Simulation standard deviations are reported in parentheses.

Our proposed estimators, **BDR** and **BDRS**, yield point estimates closest to the true values of QTEs. Notably, incorporating a shrinkage prior, as in **BDRS**, further enhances the performance of **BDR**, particularly when the object of interest is extreme tails (i.e. 10th and 90th percentiles). Despite sharing a probabilistic approach, the Bayesian nonparametric estimator, **BNP**, demonstrates differences from our proposed estimators. While the nonparametric method produces point estimates slightly better than **Naive** method, they are still far from the truth. It is worth noting that because both **BADR** estimators and **BNP** use BART-logit to model the treatment assignment in the first stage, their distinct performance illustrates the role of modeling the conditional distribution of potential outcomes given confounders. Intuitively, **BNP** avoids directly modeling the conditional distribution of potential outcomes given confounders. Instead, it is grounded in the balancing property of the propensity score (Rosenbaum and Rubin, 1983) to model the conditional distribution of the outcome given the propensity score alone. While this approach involves estimating a less complex distribution due to having only one binary regressor (i.e. the estimated propensity score) in the second stage, it becomes skeptical in the case of misspecified treatment assignment. According to Monte Carlo results in the original paper by Xu et al. (2018), the inclusion of non-confounders in the treatment assignment equation entails less precise estimations of QTEs, thereby compromising the performance of the **BNP** method. Aside from efficiency loss, finite-sample bias is also a notable drawback of methods targeting a set of variables that best predict treatment assignment without accounting for how these variables are related to the outcome, as widely discussed in the context of

average treatment effect (Zigler and Dominici, 2014; Belloni et al., 2014, etc.).

Among frequentist approaches, TMLE and FIPW perform reasonably well in this relatively favorable sample size, although they do not surpass our proposed estimators. In contrast, LDML retains somewhat larger finite-sample bias.

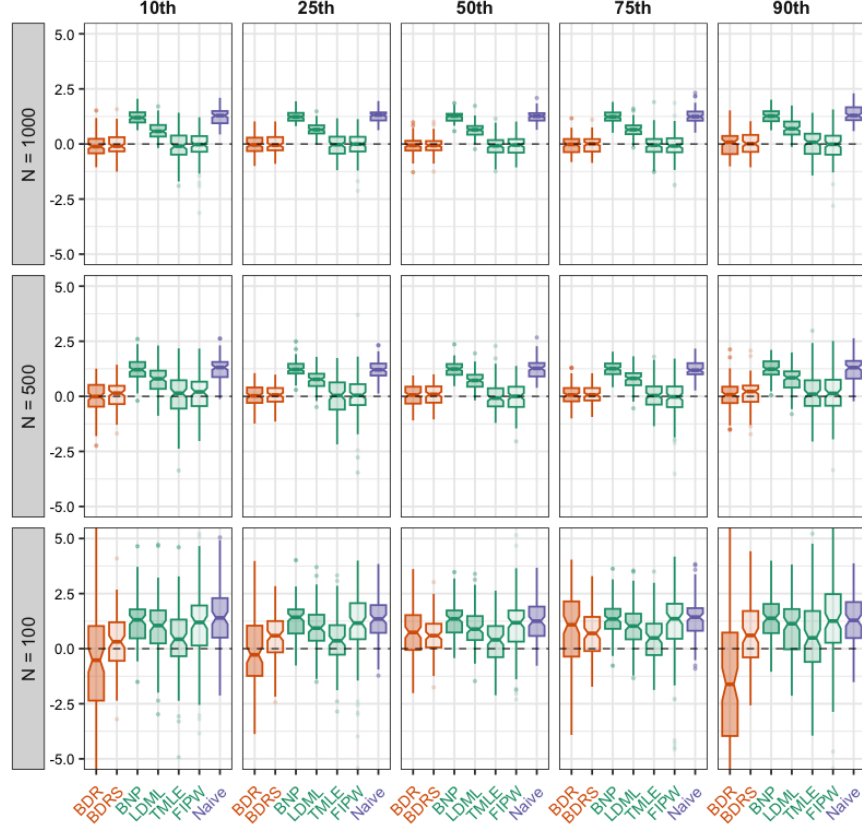


Figure 4.2: Sampling distributions of QTE estimation errors, $\hat{\Delta}_\tau - \Delta_{\tau,0}$, at 10th, 25th, 50th, 75th, and 90th percentiles across 100 replicates. The horizontal dashed line denotes zero estimation error.

Boxplots in Figure 4.2 offer more insight on the sampling distributions of the difference between the estimated and the true quantities ($\hat{\Delta}_\tau - \Delta_{\tau,0}$), which produced by all estimators across 100 simulated datasets. When $N = 1000$, BDR and BDRS showcase nearly zero median (or mean) bias as well as small variation, outperforming other methods. Their strong performance persists even in smaller sample size of $N = 500$. Interestingly, the advantage of BDRS, which is developed by adopting a hierarchical shrinkage prior, becomes prominent when $N = 100$. While the performance of BDR exhibits instability in the presence of high-dimensional covariates, BDRS handles such settings more effectively, as evidenced by the remarkably reduced box widths observed for all QTEs of interest.

Tables 4.2 and 4.3 quantify these patterns. At $N = 1000$, BDRS obtains the smallest MAE at each of the five quantiles, while BDRS has the smallest RMSE at the 10th percentile and BDRS has the

smallest RMSE at the remaining four quantiles. At $N = 500$, BDRS attains the smallest MAE and RMSE at every quantile considered. Even in the most challenging scenario, $N = 100$ and $p = 40$, BDRS has the lowest MAE and RMSE at four of the five quantiles; TMLE performs slightly better at the 75th percentile. Thus, the proposed BADR procedures consistently achieve the best overall bias-variance tradeoff across quantiles and sample sizes, and shrinkage is particularly advantageous in the small- N , high- p/N regime.

The rate at which performance improves with N is also informative. For BDR and BDRS, both MAE and RMSE fall sharply as p/N decreases. A similar pattern occurs at the other quantiles. In contrast, the errors of the Naive and BNP estimators remain comparatively persistent as N increases, indicating that their dominant problem is systematic bias rather than sampling variability. These results therefore distinguish gains from increasing sample size from gains due to appropriate confounding adjustment and targeted debiasing.

Table 4.2: Root Mean Square Error (RMSE)

Percentiles	N	Estimation Methods						
		BDR	BDRS	BNP	LDML	TMLE	FIPW	NAIVE
10th	1000	0.49	0.499	1.262	0.72	0.72	0.696	1.299
	500	0.709	0.65	1.315	0.97	1.065	1.104	1.361
	100	2.927	1.34	1.589	1.635	1.627	2.231	1.863
25th	1000	0.409	0.394	1.266	0.707	0.505	0.549	1.299
	500	0.489	0.481	1.281	0.855	0.834	1.031	1.283
	100	1.615	1.12	1.537	1.409	1.268	2.059	1.64
50th	1000	0.359	0.355	1.263	0.702	0.473	0.475	1.279
	500	0.486	0.482	1.261	0.83	0.646	0.71	1.313
	100	1.384	1.035	1.496	1.271	1.064	2.229	1.574
75th	1000	0.404	0.398	1.275	0.742	0.544	0.594	1.326
	500	0.479	0.454	1.271	0.865	0.684	0.868	1.311
	100	1.993	1.263	1.524	1.386	1.238	2.296	1.678
90th	1000	0.527	0.512	1.32	0.843	0.625	0.696	1.388
	500	0.674	0.651	1.328	0.961	0.903	0.981	1.371
	100	3.686	1.537	1.71	1.655	2	2.818	1.809

Notes: This table displays the Root Mean Squared Error (RMSE) of different estimation methods. The rows contain results for various percentile levels and for various sample size N . The RMSE is measured by $RMSE = \sqrt{R^{-1} \sum_{r=1}^R (\hat{\alpha}_r - \alpha)^2}$.

Table 4.3: Mean Absolute Error (MAE)

Percentiles	N	Estimation Methods						
		BDR	BDRS	BNP	LDML	TMLE	FIPW	NAIVE
10th	1000	0.395	0.392	1.226	0.614	0.561	0.507	1.247
	500	0.58	0.53	1.23	0.821	0.841	0.764	1.237
	100	2.226	1.077	1.332	1.327	1.246	1.769	1.534
25th	1000	0.333	0.321	1.243	0.651	0.412	0.416	1.27
	500	0.396	0.387	1.226	0.765	0.673	0.706	1.219
	100	1.278	0.895	1.326	1.155	1.007	1.672	1.415
50th	1000	0.269	0.263	1.24	0.641	0.362	0.368	1.25
	500	0.416	0.412	1.209	0.727	0.515	0.558	1.241
	100	1.111	0.831	1.308	1.029	0.855	1.66	1.335
75th	1000	0.329	0.323	1.246	0.666	0.409	0.447	1.284
	500	0.381	0.362	1.21	0.778	0.545	0.632	1.244
	100	1.617	1.016	1.334	1.151	0.963	1.787	1.458
90th	1000	0.441	0.431	1.282	0.742	0.505	0.527	1.337
	500	0.524	0.509	1.245	0.83	0.702	0.739	1.246
	100	2.998	1.248	1.462	1.379	1.524	2.165	1.522

Notes: This table displays the Mean Absolute Error (MAE) of different estimation methods. The rows contain results for various percentile levels and for various sample size N . The MAE is measured by $MAE = R^{-1} \sum_{r=1}^R |\hat{\alpha}_r - \alpha|$.

Point accuracy alone does not establish reliable inference. Figure 4.3 therefore plots empirical coverage against average 95% interval length. The horizontal dashed line indicates the nominal 95% coverage level. An attractive procedure should lie close to this line while avoiding unnecessarily wide intervals. The BADR procedures exhibit a favorable coverage–precision trade-off. For $N = 500$ and $N = 1000$, BDR and BDRS generally achieve coverage close to the nominal level while retaining comparatively moderate interval lengths. BDRS is particularly stable in the $N = 100$ design, where regularization becomes most important. Other semiparametric procedures can also obtain satisfactory coverage in several scenarios, but often with noticeably longer intervals. LDML, in particular, can achieve high coverage at the cost of relatively wide confidence intervals. By contrast, the short intervals associated with BNP and the Naive estimator frequently accompany substantial undercoverage because their intervals are centered around biased point estimates. Figure 4.3 therefore complements the MAE and RMSE results. The advantage of BADR is not simply that it produces a small point-estimation error; it combines competitive point estimation with uncertainty quantification that is much better calibrated to repeated sampling.

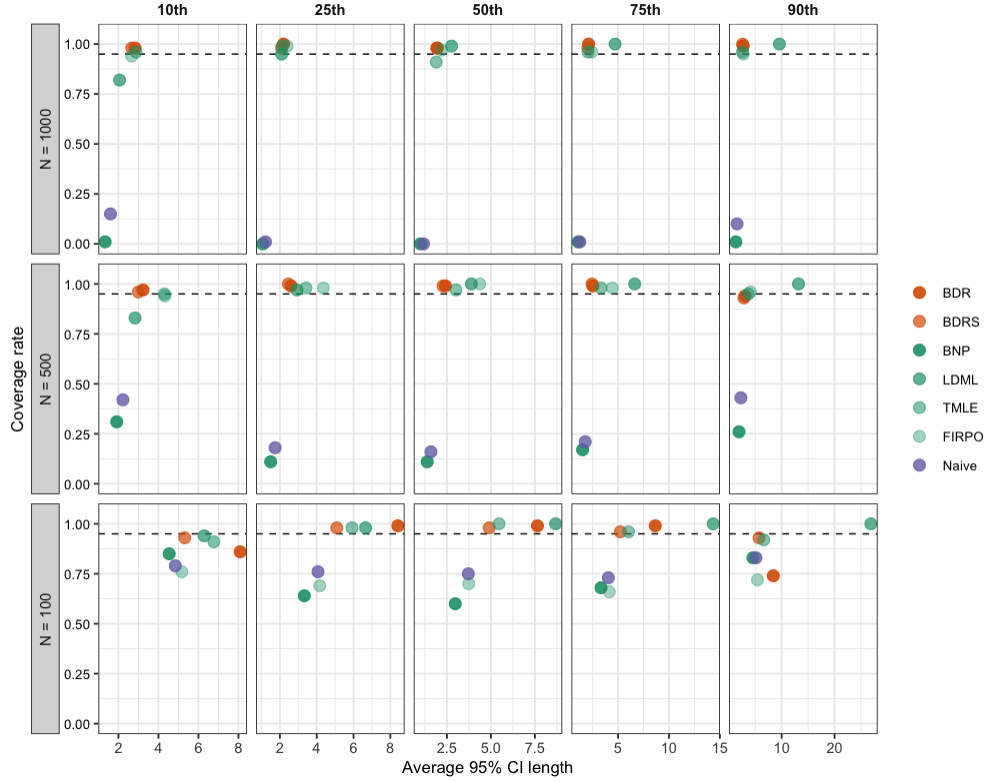


Figure 4.3: Empirical coverage probability versus Average interval lengths for 10th, 25th, 50th, 75th, and 90th QTEs across 100 replicates. The dashed horizontal line denotes the nominal 95% coverage level. Estimators closer to the dashed line and further to the left exhibit a more favorable coverage-precision trade-off.

In conclusion, BDR and BDRS demonstrate similarly excellent performance in moderate dimensionality, thereby facilitating robustness checks in practical use. BDRS even provides extra merit thanks to its adaptation to high-dimensional settings. It is trivial that the Bayesian Adaptive Lasso in BDRS is only one option among a wide range of shrinkage priors which can be incorporated into our proposed framework. Thus, the results from this simulation exercise imply the great potential of our methodology in flexibly handling high dimensions when estimating quantile treatment effects.

5 Empirical Illustration

5.1 Overview

To demonstrate the applicability and usefulness of our proposed method, we revisit the microcredit study by [Crépon et al. \(2015\)](#), which was derived from a randomized experiment conducted in Morocco. The dataset enables us to examine the potential of our approach in two distinct contexts. In the first setting, we employ the *random treatment assignment* available in the original research to investigate the effect of microcredit availability on household borrowing activities, such as the total amount of loans. Our second setting deviates from randomization – we instead use *observational data* while assuming *selection-on-observables* to evaluate the welfare impact of household loans.

The evaluation was conducted across 162 Moroccan villages that were paired based on their observable similarities. The intervention was microcredit availability, which was randomly assigned to one village within each pair. These designated villages constituted the treated group, whereas the remaining villages formed the control group. In particular, a microfinance institution was established in the treated villages between 2006 and 2007. In 2009, a follow-up study surveyed 5551 households in both treated and control villages.

The expansion of microcredit, or access to loans in general, can have potentially heterogeneous effects on household welfare for several reasons. First, households have diverse loan take-up behaviors. They may differentially select into borrowing activities based on their characteristics, leading to varying outcomes. Those who do not take up loans may end up worse off due to effects on wages or the displacement of informal lending in a dynamic general equilibrium ([Kaboski and Townsend, 2011](#); [Morduch, 1999](#)). Second, among borrowers, the effects may vary due to differences in the efficiency of loan use and uneven investment opportunities. Indeed, certain households may not benefit from loans if the requirements for investment purposes are restrictive or the term to maturity is too short ([Banerjee, 2013](#)). Additionally, multiple microlenders in a community can engage in exploitative lending practices and “overlending” to households who cannot feasibly repay the loan ([Schicks, 2013](#); [Ahmad, 2003](#)). This can result in high-productivity borrowers benefiting from the positive impact, whereas the most vulnerable borrowers are systematically harmed by the saturation of credit markets. In summary, there are potential winners and losers to financial market expansion. Even if disadvantaged groups are small, social welfare consequences could be substantial, particularly if economic inequality across households is exacerbated ([Meager, 2022](#)).

The average treatment effect (ATE), which is most commonly utilized in empirical research, cannot reveal this heterogeneity. Even though loan access might have no impact on average, it could still have significant positive or negative effects on different types of households. This policy implication is particularly critical for developing countries. To gain a more comprehensive understanding of causal effects, it is worthwhile to estimate unconditional quantile treatment effects (QTEs), which offer a valid measurement that goes beyond the ATE for the entire population. Therefore, our proposed framework is well suited for this empirical context.

In contrast to the original paper and previous studies that typically rely on randomized controlled

trial (RCT) design and ad hoc selection of baseline covariates, our approach offers more flexible specifications and data-driven estimation. This enables us to conduct new analyses using either data from randomized experiments or observational data, as demonstrated in sections 6.2 and 6.3, respectively.

Specifically, our general strategy is to initially create a large set of covariates by combining village pair dummies and full cubic polynomials along with interaction terms of household observed characteristics. Once collinear columns are removed, this set serves as the baseline specification of $\mathbf{X}$ and can be readily integrated into our Bayesian Analogue of Doubly Robust (BADR) estimation framework. Given the high dimensionality of this empirical issue, we opt for the Bayesian Doubly Robust estimator with Adaptive Lasso (BDRS) due to its proven merits in our prior simulation study. To compare our results with the benchmark, we also include the Naive estimator (**Naive**) in our analysis.

5.2 Impact of Microcredit Availability on Loan Amount

We begin with the context of random treatment assignment, where our objective of interest is the effect of microcredit availability on the total amount of loans at the household level. To examine the balance between the treated and control groups, we select pre-treatment covariates which are observed characteristics for each household, including head age, education of the head, number of adults, total number of members in a household, indicators for households doing animal husbandry, doing other non-agricultural activities, and whether household spouse responded to the survey. Table 5.1 reports the mean values of these covariates in addition to the outcome and treatment variables, both for the whole sample and for each of the treated and control groups.

Table 5.1: Summary Statistics of Households

Treatment: Microcredit Availability (RCT)	Treated	Control	Treated – Control		
	Mean (sd)	Mean (sd)	Diff.Mean	t-statistic	
Outcome variable					
Total amount of loans (in MAD)	2350.44 (10865.84)	1547.75 (7381.73)	802.69	*	2.54
Baseline covariates					
Head age	49.18 (15.83)	48.14 (15.85)	1.05	.	1.95
Head with no education	0.67 (0.47)	0.68 (0.47)	-0.01		-0.89
Number of members	5.70 (2.54)	5.64 (2.44)	0.06		0.71
Number of adults	3.81 (1.99)	3.76 (1.91)	0.05		0.83
Number of members aged 6-16	1.22 (1.29)	1.25 (1.26)	-0.03		-0.72
Declared animal husbandry activities	0.60 (0.49)	0.55 (0.50)	0.05	**	2.73
Declared non-agricultural activities	0.17 (0.37)	0.21 (0.41)	-0.04	**	-3.15
Spouse of head responded	0.09 (0.29)	0.07 (0.26)	0.02	*	2.27
Member responded	0.05 (0.22)	0.05 (0.21)	0.00		0.62

Data sources: Moroccan household survey (Crépon et al., 2015).

Although the randomization of microcredit availability and the absence of confounding factors leading to self-selection into treatment is plausible, there are slightly imbalances in covariates across

the two groups. Regarding the unconditional means, the households' total loan amount for the treated group (2350.44) significantly exceeds that of the control group (1547.75). The potential heterogeneity of microcredit motivates us to further investigate this positive average treatment effect using quantile analysis.

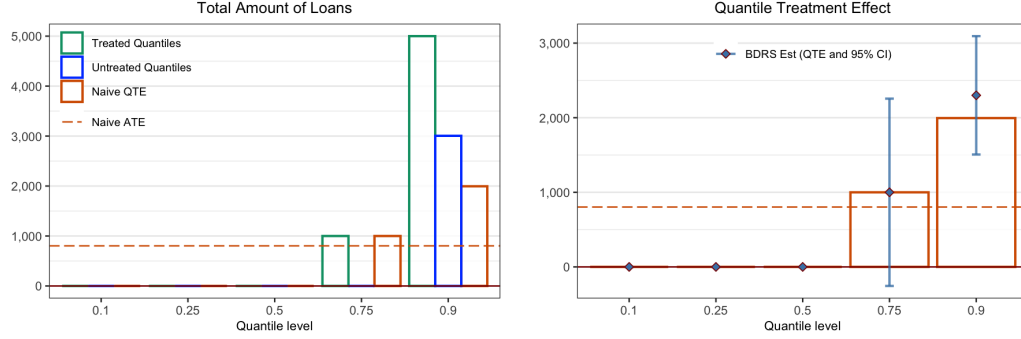


Figure 5.1: Quantile Treatment Effects (QTEs) of microcredit availability on households' total loan amount. The graph on the left demonstrates Naive estimation results. Red bar plots represent naive QTEs, which are differentials between empirical quantiles of treated group (in green) and control group (in blue). Red dashed line indicates naive Average Treatment Effects (ATE), which is simple mean difference between these two groups. Results obtained using BDRS method, QTE point estimates and corresponding 95% CI at five quantile levels based on 100 bootstrap replications, are plotted as error bars in the right-hand graph.

The results of Quantile Treatment Effects (QTEs), as estimated by the Naive and BDRS methods, are depicted in Figure 5.1. According to the findings, microcredit expansion has a precise zero effect below the 75th percentile of the distribution of total loan amounts, but exhibits positive effects above this threshold. In particular, at the 90th percentile, the positive effect is statistically significant (2300), contributing to the decomposition of the average treatment effect (802.69). Compared with naive estimates, BDRS produces similar results, only higher at the 90th percentile; however, the difference is insignificant. The result is robust after adjusting for the influence of covariate imbalance on the outcome.

Using the same dataset, findings in Chernozhukov et al. (2017) and Jacob (2021) also document the heterogeneity of the microcredit availability on total loan amount; however, their estimand is conditional ATE, different from this paper (QTEs).

5.3 Impact of Loan Access on Household Outcomes

Our second objective is to explore the causal impact of access to loans on household welfare, with a focus on the distribution of consumption and business outcomes, including total consumption, consumption of temptation goods, total output, and total profit. Unconditional QTEs provide deeper insights into the potential heterogeneity of causal effects across the distribution of each outcome interest, as well as the resulting change in household inequality. The binary treatment we consider is the actual borrowing status recorded at the household level. Table 5.2 indicates that the difference in mean between two groups of households (borrowers and non-borrowers) is highly statistically significant regarding consumption, but not for business outcomes. However, there are

two caveats to these naive ATE estimates.

Table 5.2: Summary Statistics of Household Outcomes.

Outcome variables	Borrowers		Non-borrowers		Borrowers – Non-borrowers	
	Mean	St.Dev.	Mean	St.Dev.	Diff.Mean	t-statistic
<i>(in MAD)</i>						
Total Consumption	3268.62	(2956.01)	2863.49	(1792.97)	405.13 ***	3.82
Temptation Goods	312.33	(229.91)	270.31	(219.33)	42.01 ***	4.73
Total Output	32672.06	(85071.58)	30885.38	(85939.63)	1786.68	0.54
Total Profit	10081.86	(37986.07)	8409.95	(45277.88)	1671.91	1.07

Data sources: Moroccan household survey (Crépon et al., 2015). *Definition:* Total Consumption is monthly total consumption (in MAD); Temptation Goods is monthly expenditure on temptation and entertainment(in MAD); Total Output is sum of agricultural, livestock, and non-agricultural business production over the 12 months prior to the survey (in MAD); Total Profit is total profit of self-employment activities over the 12 months prior to the survey (in MAD).

Firstly, all outcome variables in this empirical setting exhibit heavy tails and large variability, as illustrated in histograms in Figure 5.2. This is another motivation for quantile analysis since estimation results for a set of quantiles would be less susceptible to the influence of outliers than results for the mean.

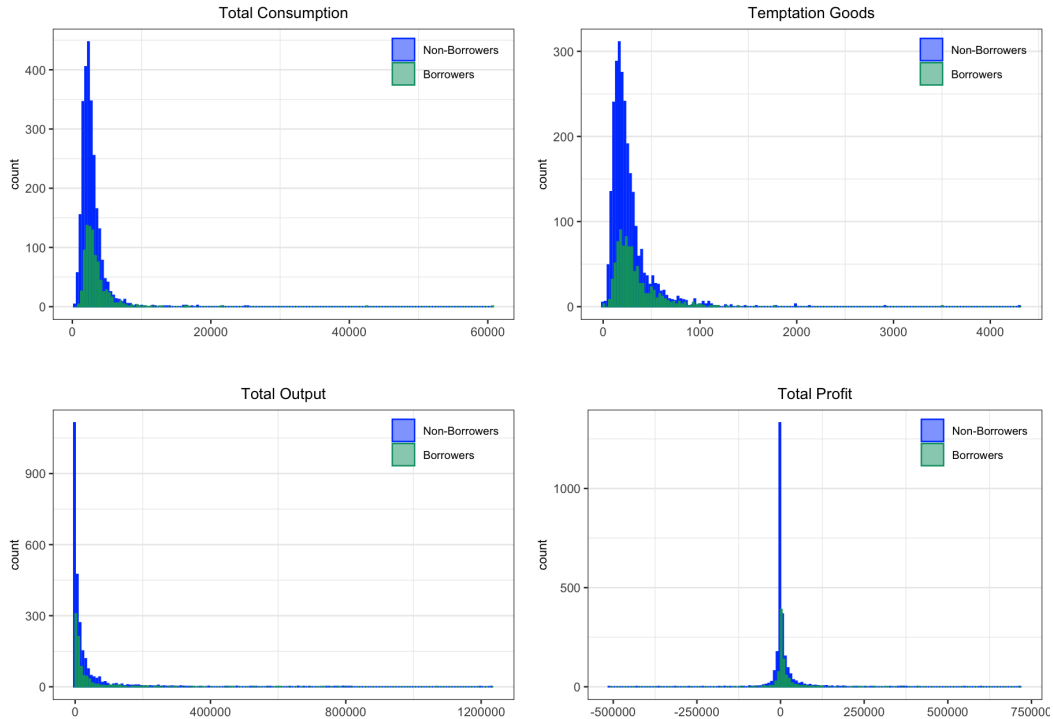


Figure 5.2: Histograms of various consumption and business outcomes of borrowing households (in green) and nonborrowing households (in blue). These graphs display raw data without any truncation applied.

Secondly, the treatment variable – borrowing pattern observed in the dataset – is no longer

randomly assigned among households in the present context. This is confirmed by the imbalances between these two groups regarding the mean values of the observed characteristics, as shown in Table 5.3. Specifically, borrowing households tend to have larger average household sizes. They are also more inclined to engage in non-agricultural self-employment activities and reside in villages where microcredit is available. The discrepancy observed is more than what would be expected by pure chance. Therefore, to identify causal effects using non-experimental data, we pursue the *selection-on-observables* assumption. That means, conditional on observed covariates, unmeasured factors that influence household loan access are independent of household outcomes.

Table 5.3: Covariate Balance between Borrowers and Non-borrowers.

Control variables	Borrowers	Non-borrowers	Borrowers – Non-borrowers	
	Mean (sd)	Mean (sd)	Diff.Mean	t-statistic
Head age	49.01 (15.62)	48.53 (15.93)	0.49	0.79
Head with no education	0.68 (0.47)	0.68 (0.47)	0.00	0.05
Number of members	6.06 (2.46)	5.54 (2.48)	0.52	*** 5.36
Number of adults	4.02 (2.01)	3.71 (1.92)	0.31	*** 3.99
Number of members aged 6-16	1.36 (1.30)	1.19 (1.27)	0.16	** 3.23
Declared animal husbandry activities	0.59 (0.49)	0.57 (0.50)	0.02	1.23
Declared non-agricultural activities	0.23 (0.42)	0.18 (0.38)	0.05	** 3.17
Spouse of head responded	0.05 (0.23)	0.09 (0.29)	-0.04	*** -3.96
Member responded	0.05 (0.21)	0.05 (0.22)	0.00	-0.36
Microcredit availability	0.55 (0.50)	0.47 (0.50)	0.07	*** 3.73

Data sources: Moroccan household survey (Crépon et al., 2015).

Whilst a violation of randomization may threaten the performance of the naive estimator, the BDRS estimator serves as a debiasing device, as illustrated in our simulation using synthetic data. Table 5.4 presents the results for key outcome variables related to household consumption and business. Unlike the first setting, the BDRS estimates differ considerably from the naive estimates of QTEs because selection bias is accounted for in our proposed approach. Overall, the point estimates at extreme tails (10th and 90th percentiles) are fairly imprecise, as indicated by large credible intervals compared to the other quantile levels. Interestingly, the causal effect in the upper tail remains significantly positive in most cases.

Regarding *total consumption*, although all naive estimates of ATE and QTE are positive, estimation results obtained using the BDRS method reveal notably lower effects across all quantile levels. The effects of loan access are most pronounced at the 75th and 90th percentiles of the consumption distribution; however, they are insignificantly positive. While the naive method overestimates the effect of loan access compared to the BDRS estimator, the upward bias suggests a possible selection-on-gain pattern. Households inclined to borrow to support their consumption are more likely to gain higher total consumption when they have financial access.

Further examination of the impact of borrowing on *temptation consumption* shows a similar upward bias in the Naive method relative to the BDRS method. The effect is slightly negative at the

lowest percentile (10th) yet clearly insignificant. By contrast, significant positive effects are observed at the median and higher percentiles. This seems inconsistent with other works that have found a statistically significant reduction in nonessential expenditures. However, these studies used different treatment variables and designs compared to this paper.

Table 5.4: Quantile Treatment Effects of Loan Access on Household Outcomes.

Outcomes	Percentiles	BDRS			Naive
		QTEs	Upper bound	Lower bound	QTEs
Total Consumption	10th	20.093	1469.207	-1429.020	232.795
	25th	9.242	173.511	-155.027	173.456
	50th	79.949	229.587	-69.690	229.753
	75th	132.22	273.974	-9.534	286.680
	90th	237.699	543.355	-67.956	685.442
Temptation Goods	10th	-8.69	65.660	-83.040	17.380
	25th	13.035	29.871	-3.801	21.725
	50th	30.415	45.962	14.868	43.450
	75th	47.795	79.225	16.365	60.830
	90th	78.21	129.145	27.275	78.210
Total Output	10th	0	13146.948	-13146.948	0.000
	25th	-330	19.769	-679.769	1093.446
	50th	50	1385.992	-1285.992	1787.500
	75th	1666	6933.205	-3601.205	2771.616
	90th	27360	52964.198	1755.802	2744.044
Total Profit	10th	-5500	-1183.536	-9816.464	-1142.697
	25th	-945	158.825	-2048.825	-241.876
	50th	561	1117.727	4.273	979.125
	75th	1780.769	4273.915	-712.377	549.373
	90th	8954.377	16664.233	1244.520	-1086.350

Notes: Upper bound and Lower bound for BDRS method are associated with the estimates of 95% CI based on 100 bootstrap replications.

With respect to household business outcomes, there is a statistically significant increase in *total output*, concentrated only at the highest quantile level. For the rest of the community, specifically those below the 90th percentile, no systematic change appears to be taking place. Consequently, the majority of the total output distribution remains unchanged with or without universal access to loans.

Interestingly, there is notable evidence of heterogeneous effects on *total profit*. The effect on the median household estimated by the BDRS method is quite close to the naive ATE estimate, which is moderately positive. In general, access to loans has a favorable impact on households' profit by shifting the center of distribution towards the right. However, the impacts exhibited at extreme tails are more dramatic, with a negative effect at the lowest percentile (10th) and a positive effect at the highest percentile (90th), and both are statistically significant. If the rank invariance assumption is invoked, the rightward expansion of the upper tail means that high-profit households gain benefits,

while the leftward expansion of the lower tail means that low-profit households experience loss when loans are accessible to everyone, compared to the opposite counterfactual scenario. While this assumption might be difficult to defend given the complexity and nonlinearities inherent to the financial market, interpretations about the shape change of distribution of household total profit remain valid. There do exist both winners and losers, even when we cannot identify the specific households that belong to each group. The outcome distribution disperses wider leading to the exacerbation of inequality across households.

Taken together, the estimated QTE patterns collectively indicate that broadening financial access is likely to result in an ex-post rise in economic inequality across households. Specifically, the increase in total output and consumption at the household level is solely attributable to the right tail of distributions expanding rightward, suggesting that certain households are likely to experience an improvement in their economic circumstances without incurring any systematic losses from others. Further investigation of total profit, however, reveals a more nuanced picture. While the overall impact of access to loans on total profit is positive, indicating a shift towards higher profits for many households, there is evidence of extreme heterogeneity. The effect is asymmetric as certain households may experience negative effects on their profits.

Although the treatment variable and identification strategies employed in this setting differ from those used in [Crépon et al. \(2015\)](#), the findings converge in several respects. In the original paper, both the reduced-form quantile regressions and instrumental variable (IV) estimates suggest substantial heterogeneity in the profitability of microfinance investments and emphasize the detrimental effects on certain households. Specifically, their reduced-form quantile analysis measures Intention-to-Treat (ITT) effects because the treatment variable is microcredit availability at the village level rather than actual borrowing at the household level. Additionally, the IV estimates in this study reveal changes in the unconditional distribution of total profit for those who take up microcredit (i.e., compliers only). These results are only valid when randomization holds. In contrast, the findings of this paper have broader implications for the understanding of economic inequality, as we focus on the entire population of households utilizing non-experimental data and a *selection-on-observables* assumption.

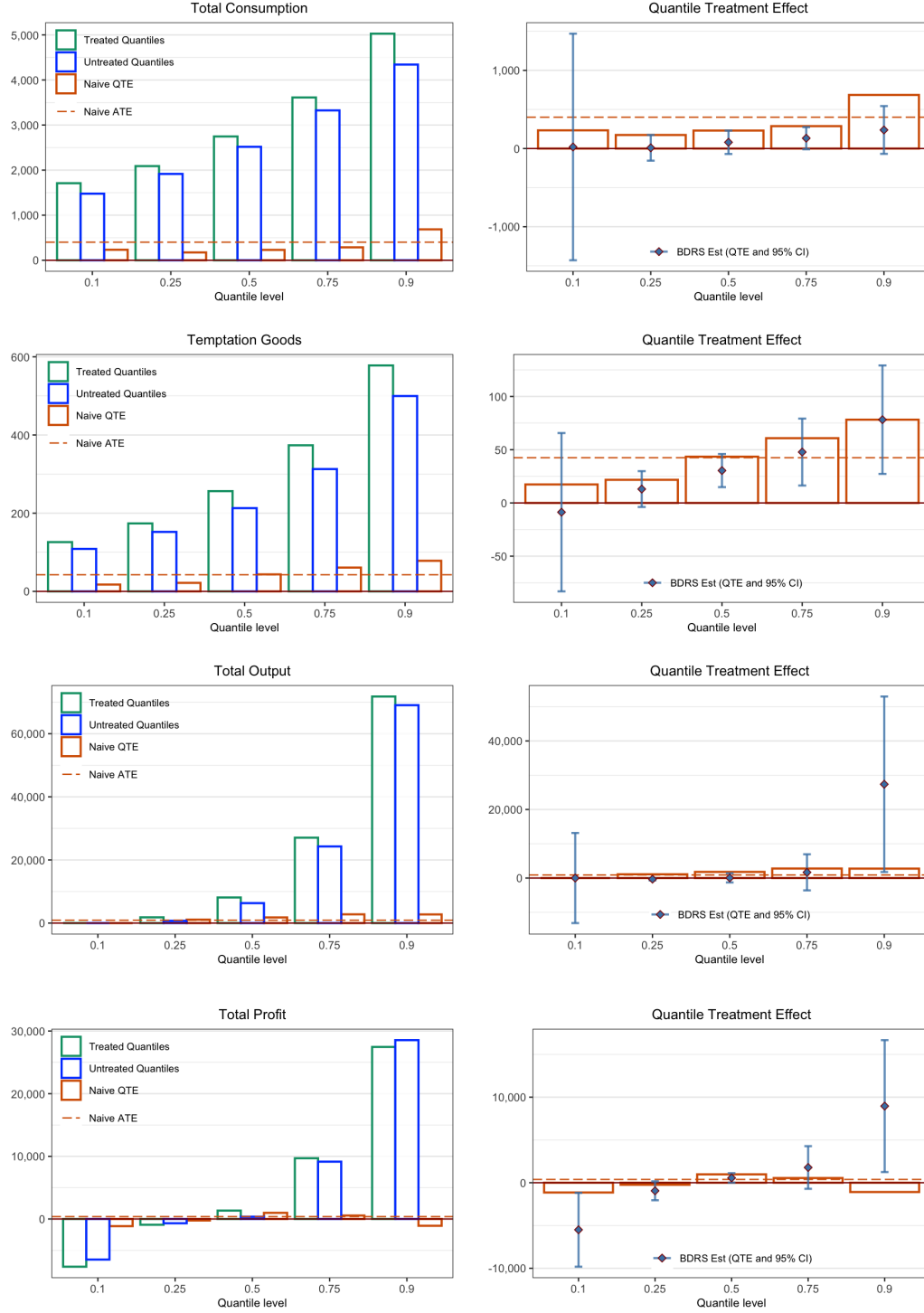


Figure 5.3: Quantile Treatment Effects (QTEs) of loan access on various household outcomes. Graph on the left demonstrates Naive estimation results. Red bar plots represent naive QTEs, which are differentials between empirical quantiles among borrowing households (in green) and nonborrowing households (in blue). Red dashed line indicates naive Average Treatment Effect (ATE), which is simple mean difference between these two groups. Results obtained using BDRS method, QTE point estimates and corresponding 95% CI at five quantile levels based on 100 bootstrap replications, are plotted as error bars in the right-hand graphs.

6 Conclusion

This paper develops a Bayesian Analog of Doubly Robust (BADR) framework for estimation and inference on unconditional Quantile Treatment Effects (QTEs) in observational studies with potentially high-dimensional covariates. The proposed approach combines flexible Bayesian learning of the propensity score and conditional outcome distribution with a target-specific, one-step posterior correction based on the efficient influence function. Bayesian-bootstrap perturbations are incorporated into the correction to account for the first-order sampling uncertainty of the QTE functional. In this way, the framework is designed to retain the regularization and predictive flexibility of modern Bayesian methods while providing inference that is calibrated from a semiparametric frequentist perspective.

We establish a semiparametric Bernstein–von Mises theorem for the resulting BADR-QTE posterior. Under posterior concentration, quantile regularity, nuisance-contraction, and influence-function stability conditions, the corrected posterior is asymptotically Gaussian around an efficient estimator with variance equal to the semiparametric efficiency bound. Consequently, posterior means are asymptotically efficient and credible intervals attain asymptotically correct frequentist coverage. Importantly, the required nuisance-rate condition takes a product form, allowing slower convergence of the propensity score to be compensated by faster convergence of the conditional outcome distribution, and vice versa. The procedure therefore inherits a rate double-robustness property while permitting flexible, potentially high-dimensional nuisance estimation.

The Monte Carlo study provides complementary finite-sample evidence. Across different sample sizes and quantile levels, the proposed BADR estimators generally deliver competitive or lower estimation error along with well-calibrated uncertainty quantification relative to existing Bayesian and frequentist alternatives. The gains from shrinkage are particularly pronounced when the dimension of the covariate vector is large relative to the sample size. Moreover, the posterior-correction exercises highlight an important distinction between nuisance learning and inference on the target functional. An uncorrected plug-in posterior can remain overly concentrated even when its point estimate is reasonably accurate. The efficient-influence-function correction improves posterior centering, while the Bayesian-bootstrap perturbation restores the uncertainty required for substantially improved coverage. These findings reinforce the theoretical motivation for correcting the posterior rather than relying directly on the plug-in posterior induced by flexible Bayesian nuisance models.

We further illustrate the proposed framework by revisiting the effects of access to microcredit on household consumption and business outcomes. The empirical results demonstrate the value of moving beyond average treatment effects when interventions have heterogeneous distributional consequences. The estimated QTEs reveal effects that vary considerably across different parts of the outcome distribution and provide evidence of changes in the dispersion of household outcomes that would be obscured by an analysis focused exclusively on means. In particular, the distributional analysis indicates that access to credit may generate substantial gains for some households while adverse effects on business profits remain possible for others, underscoring the relevance of QTEs

for welfare and policy analysis.

Several extensions remain open. First, it would be useful to verify the high-level conditions underlying the semiparametric Bernstein–von Mises theorem for broader classes of Bayesian nonparametric and machine-learning priors. Second, the posterior-correction principle developed here may be extended to other nonlinear semiparametric causal functionals for which nuisance parameters are high-dimensional or depend on the target parameter itself. Finally, the current analysis relies on selection on observables for identification. Extending the BADR framework to settings with endogenous treatment assignment driven by unmeasured confounding would substantially broaden its applicability.

APPENDIX

A Proofs of Main Results

A.1 Proof of Proposition 3.1

Proof. For $q = q_{t,\theta,n}$, iterated expectations give

$$P_0\{\phi_{t,\theta,n} - \tau\} = F_{t0}(q) - \tau + B_t(\theta, q),$$

where

$$B_t(\theta, q) = P_0 \left[\left\{ \frac{\pi_{t0}(X)}{\pi_t(X)} - 1 \right\} \{G_{t0}(q | X) - G_t(q | X)\} \right].$$

Positivity and Cauchy–Schwarz imply

$$|B_t(\theta, q)| \lesssim \rho_n \epsilon_n$$

uniformly over $\mathcal{H}_n^\theta$. Because $F_{t0}(q_{t0}) = \tau$, Taylor expansion around q_{t0} yields

$$F_{t0}(q) - \tau = f_{t0}(q_{t0})(q - q_{t0}) + O(a_n^2).$$

Using

$$P_0\varphi_{t,\theta,n} = -\frac{F_{t0}(q) - \tau + B_t(\theta, q)}{d_{t,\theta,n}},$$

together with

$$|d_{t,\theta,n} - f_{t0}(q_{t0})| \leq b_n$$

and the uniform lower bound on $d_{t,\theta,n}$, gives

$$|r_{2,t}(P_0, \theta)| \lesssim a_n^2 + a_n b_n + \rho_n \epsilon_n.$$

The scaled conclusion follows from (3.6). □

B Bayesian Additive Regression Tree (BART)

B.1 BART Model Specifications

BART is a nonparametric modeling technique that translates decision tree-based ensemble methods to a Bayesian framework. [Chipman et al. \(2010\)](#) present a comprehensive overview of the method. In essence, BART is a *sum-of-trees* model with prior distributions are placed over the parameters including *tree depth*, *splitting variables*, *splitting values*, and *terminal node* estimates.

Consider the regression problem that predicts a continuous Y_i using a p -dimensional vector of predictors $\mathbf{X}_i = (X_{i1}, \dots, X_{ip})^\top (i = 1, \dots, N)$, BART model can be expressed as

$$Y_i = f_{\text{BART}}(\mathbf{X}_i) + \epsilon_i, \quad \epsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2), \quad f_{\text{BART}}(\mathbf{X}_i) = \sum_{m=1}^M f_{\text{tree}}(\mathbf{X}_i; \Gamma_m, \mu_m), \quad (\text{B.1})$$

where $f_{\text{tree}}(\mathbf{X}_i; \Gamma_m, \mu_m)$ is a Bayesian single regression tree; Γ_m is a tree structure that consists of a set of splitting rules and a set of terminal nodes; and $\mu = (\mu_{m,1}, \dots, \mu_{m,b_m})$ is a vector of parameters associated with b_m terminal nodes of Γ_m , such that $f_{\text{tree}}(\mathbf{X}_i; \Gamma_m, \mu_m) = \mu_{m,l}$ if $\mathbf{X}_i$ is corresponding to the l^{th} terminal node of Γ_m .

The prior of BART is specified for three components:

1. The ensemble structure $\{\Gamma_m\}_{m=1}^M$

Independent regularization prior is placed on Γ_m . It consists of a Bernoulli distribution with probability

$$\Pr(\text{split} \mid d) = \alpha(1 + d)^{-\beta}, \quad \alpha \in (0, 1), \quad \beta \in (0, \infty), \quad (\text{B.2})$$

for splitting a node at tree depth d ($d = \{0, 1, \dots\}$) into two child nodes and two discrete uniform distributions for selecting a split variable and a split value given the selected split variable. This regularization prior helps prevent individuals from becoming too influential, thereby enhancing the overall fit and mitigating the risk of overfitting.

2. The parameters $\{\mu_m\}_{m=1}^M$ associated with the terminal nodes given $\{\Gamma_m\}_{m=1}^M$

$$\mu_{m,l} \stackrel{iid}{\sim} \mathcal{N}(0, v) \quad (\text{B.3})$$

3. The error variance σ^2 that is independent with the former two

$$\sigma^2 \sim \text{Inv-Gamma}(r, s) \quad (\text{B.4})$$

The process of sampling the posterior distribution is carried out using a Metropolis-within-Gibbs MCMC sampler, which can also be regarded as a special case of (generalized) Bayesian backfitting algorithm ([Hastie and Tibshirani, 2000](#)), to update each tree iteratively. Estimated outcome is achieved by averaging the posterior samples of $f_{\text{BART}}(\mathbf{X}_i)$ after a burn-in period.

For binary outcome, the continuous BART model above has been extended to probit BART and

logit BART, which are specified as follows

$$\mathbb{P}(Y_i = 1 \mid \mathbf{X}_i) = H[f_{\text{BART}}(\mathbf{X}_i)], \quad (\text{B.5})$$

where $f_{\text{BART}}(\mathbf{X}_i)$ is the sum-of-trees function in (B.1) and H is the link function with the probit link for probit BART and the logit link for logit BART. Both of the models maintain the same prior assigned to the ensemble structure and the parameters of the terminal nodes, i.e. $\{\Gamma_m, \mu_m\}_{m=1}^M$, but σ^2 is fixed for the sake of identifiability.

Probit BART employs data augmentation of [Albert and Chib \(1993\)](#) to adapt the Bayesian backfitting sampler used in continuous BART. This involves introducing a latent variable Y_i^* such that $Y_i = \mathbb{1}\{Y_i^* > 0\}$ for each response variable Y_i . At each iteration of the MCMC algorithm, Y_i^* is imputed by sampling from the full conditional distribution of Y_i^* given Y_i and other parameters, which is essentially a truncated normal distribution. The imputed Y_i^* 's are then modelled using the continuous BART model with σ^2 set to 1, enabling the completion of the MCMC algorithm by performing the Bayesian backfitting algorithm of the continuous BART model on the imputed Y_i^* 's.

Logit BART also introduces latent variable Y_i^* that is instead assumed to follow a logistic distribution, which has a heavier tail than a normal distribution, thus improving estimation for extreme instances of $\mathbb{P}(Y_i = 1 \mid \mathbf{X}_i)$. These latent Y_i^* 's are sampled using the method described by [Gramacy and Polson \(2012\)](#). Conditional on the imputed Y_i^* 's, the continuous BART model with given heteroskedastic variance σ_i^2 's are fitted on Y_i^* 's, where σ_i^2 's are obtained through the technique outlined by [Robert \(1995\)](#).

B.2 On Implementation

Computational details and implementation using **BART** R package can be found in [Sparapani et al. \(2021\)](#). The actual sampling and computation are carried out in C++ code to maximize computational efficiency. Both options for Probit BART and Logit BART are available in this package, which can be utilised directly to fit the treatment assignment model in our proposed BADR framework.

C Bayesian Shrinkage Priors

C.1 Hierarchical Bayes for Linear Regression

Consider the linear regression model

$$Y_i = \mathbf{X}_i\beta + \epsilon_i, \quad \epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2) \quad (\text{C.1})$$

Assuming that interest lies in learning about the regression coefficients β , then a simple hierarchical specification takes the form

$$\begin{aligned} Y_i \mid \beta, \sigma^2 &\sim N(\mathbf{X}_i\beta, \sigma^2), i = 1 \dots, n \\ \beta_j \mid \tau^2, \sigma^2 &\sim N(0, \sigma^2\tau^2), i = 1 \dots, p \\ \tau^2 &\sim F(a, b) \\ \sigma^2 &\sim \frac{1}{\sigma^2} \end{aligned} \quad (\text{C.2})$$

where $F(a, b)$ denotes some distribution function with hyper-parameters a, b . Due to the fact that choice of τ^2 is so crucial for the posterior outcome of β_j , the idea behind this hierarchical specification is to treat the hyper-parameter τ^2 as a random variable and learn about it via Bayes Theorem.

C.2 Bayesian Shrinkage Priors

The major goal of shrinkage priors is to shrink small coefficients to zero while maintaining true large coefficients, especially in high-dimensional settings. The possible variation in shrinkage amounts among those priors depends on their specific designs. In particular, the sharper the peak is around zero, the stronger shrinkage for small coefficients. Also, the heavier the tail, the lighter the shrinkage for large coefficients.

Bayesian Lasso

The Bayesian counterpart of the Lasso penalty is Laplace prior, which was first proposed by [Park and Casella \(2008\)](#). The Bayesian Lasso can be obtained as a scale mixture of a Normal density with an Exponential density as below:

$$\begin{aligned} \beta_j \mid \tau_j^2, \sigma^2 &\sim N(0, \sigma^2\tau_j^2) \\ \tau_j^2 \mid \lambda^2 &\sim \text{Exp}\left(\frac{\lambda^2}{2}\right), \text{ for } j = 1, \dots, p \\ \lambda &\sim \text{half-Cauchy}(0, 1) \\ \sigma^2 &\sim \frac{1}{\sigma^2} \end{aligned} \quad (\text{C.3})$$

Integrating τ_j^2 out leads to Double-exponential⁸ or Laplace priors on the regression coefficients, i.e.,

$$\beta_j \mid \lambda, \sigma \sim \text{Double-exponential} \left(0, \frac{\sigma}{\lambda} \right), \text{ for } j = 1, \dots, p \quad (\text{C.4})$$

Although this version of Bayesian Lasso is the most popular form in literature so far; there are also some alternative formulations suggested by [Hans \(2009\)](#), [Mallick and Yi \(2014\)](#) and [Alhamzawi and Taha Mohammad Ali \(2020\)](#).

In addition to the overall shrinkage parameter λ , the Lasso prior has an additional predictor-specific shrinkage parameter τ_j . Therefore, the Lasso prior is more flexible than the Ridge prior, which only relies on the overall shrinkage parameter.

Horseshoe prior

A novel shrinkage prior in the Bayesian literature is the horseshoe prior [Carvalho et al. \(2010\)](#)⁹. This prior is particularly attractive for sparse signal recovery.

$$\begin{aligned} \beta_j \mid \tau_j^2 &\sim N(0, \tau_j^2) \\ \tau_j \mid \lambda &\sim \text{half-Cauchy}(0, \lambda), \text{ for } j = 1, \dots, p \\ \lambda \mid \sigma &\sim \text{half-Cauchy}(0, \sigma) \end{aligned} \quad (\text{C.5})$$

The half-Cauchy prior can be written as a mixture of inverse Gamma densities¹⁰ ([Makalic and Schmidt, 2015](#)), so that the horseshoe prior in Equation (C.5) can be equivalently specified as:

$$\begin{aligned} \beta_j \mid \tau_j^2 &\sim N(0, \tau_j^2) \\ \tau_j^2 \mid \omega &\sim \text{Inv-Gamma} \left(\frac{1}{2}, \frac{1}{\omega} \right) \\ \omega \mid \lambda^2 &\sim \text{Inv-Gamma} \left(\frac{1}{2}, \frac{1}{\lambda^2} \right) \\ \lambda^2 \mid \gamma &\sim \text{Inv-Gamma} \left(\frac{1}{2}, \frac{1}{\gamma} \right) \\ \gamma \mid \sigma^2 &\sim \text{Inv-Gamma} \left(\frac{1}{2}, \frac{1}{\sigma^2} \right) \end{aligned} \quad (\text{C.6})$$

⁸Mathematical representation:

$$\int_0^\infty \frac{1}{\sqrt{2\pi\sigma^2 s_j}} e^{\left(-\frac{\beta_j^2}{2\sigma^2 s_j}\right)} \frac{\lambda^2}{2} e^{-\frac{\lambda}{2s_j}} ds_j = \frac{\lambda}{2\sqrt{\sigma^2}} e^{-\lambda|\beta_j|/\sqrt{\sigma^2}}$$

⁹Note that [Carvalho et al. \(2010\)](#) explicitly include the half-Cauchy prior for λ in their specification, thereby implying a full Bayes approach. This formulation results in a horseshoe prior that is automatically scaled by the error standard deviation σ .

¹⁰If $x^2 \mid z \sim \text{inv-Gamma}(1/2, 1/z)$ and $z \sim \text{inv-Gamma}(1/2, 1/\alpha^2)$ then $x \sim C^+(0, \alpha)$

An expression for the marginal prior of the regression coefficients β_j is not analytically tractable, but a tight lower bound [Carvalho et al. \(2010\)](#) can be used instead.

$$-\log p(\beta_i | \lambda) \geq -\log \log \left(1 + \frac{2\lambda^2}{\beta_j^2} \right) \quad (\text{C.7})$$

The key features for the appealing performance of horseshoe prior are its Cauchy-like tails and an asymptote at origin (unique advantage), which make horseshoe adaptive to sparsity and robust to large signals so outperform other shrinkage priors we have discussed.

In the search for intuitive reasons, we consider a common framework of shrinkage rules. Define $\kappa_j = 1/(1 + \tau_j^2)$, then κ_j is a random shrinkage coefficient in $[0, 1]$. Under a multivariate normal scale mixture prior (i.e. the general form of all shrinkage priors we are discussing), the posterior mean can be written as a linear function of the observation:

$$\mathbb{E}[\beta_j | Y_j] = \{1 - \mathbb{E}[\kappa_i | Y_j]\}Y_j \quad (\text{C.8})$$

Hence, $\mathbb{E}[\kappa_i | Y_j]$ implies the amount of weight that the posterior mean for β_j places on 0 once the data have been observed. A shrinkage coefficient κ_j that is close to zero leads to virtually no shrinkage, thus describes signals. A shrinkage coefficient κ_j that is close to one leads to nearly-total shrinkage, thus describes noises. Intuitively speaking, the behavior of a priori $p(\kappa_j)$ near $\kappa_j = 1$ will control the robustness of signal at tail, while near $\kappa_j = 0$ will control the shrinkage of noise toward 0. Because of difference choice of $p(\tau_j)$, each type of shrinkage prior has distinct $p(\kappa_j)$ reflecting its attempt to separate signal and noise. For horseshoe prior, the attempt is even implied in its name, which arises from the fact that for fixed values $\lambda = \sigma = 1$, $p(\kappa_j)$ is similar to a horseshoe-shaped

Beta $(1/2, 1/2)$. This prior is symmetric and unbounded at both 0 and 1; thereby, small coefficients (noises) are heavily shrunk towards zero while substantial coefficients (signals) remain large. None of these common shrinkage priors above shares this characteristic. For instance, the Laplace prior, where $p(\kappa_j)$ is bounded at both 0 and 1, tends to over-shrink strong signals yet under-shrink noises. [Carvalho et al. \(2009\)](#), [Carvalho et al. \(2010\)](#) provide more explanation for other priors.

In fact, unlike local shrinkage priors above, the horseshoe prior is a member of a wider class of *global-local shrinkage priors* ([Polson and Scott, 2010](#); [Bhadra et al., 2019](#)) because it enables a clear separation between global and local shrinkage effects. Put another way, this class of priors adapt to sparsity by a global shrinkage parameter and recover signals by a local shrinkage parameter.

D Bayesian Quantile Regression (BQR)

D.1 Bayesian Quantile Regression

Consider the linear quantile regression model (Koenker and Bassett Jr, 1978) at a given quantile level $\tau \in (0, 1)$

$$\mathcal{Q}_\tau(Y | \mathbf{X}) = \mathbf{X}\beta_{(\tau)}, \quad (\text{D.1})$$

The quantile specific coefficient $\beta_{(\tau)}$ can be consistently estimated by

$$\hat{\beta}_{(\tau)} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^n \rho_\tau(Y_i - \mathbf{X}_i\beta), \quad (\text{D.2})$$

where $\rho_\tau(u) = u(\tau - \mathbb{1}\{u < 0\})$ is the quantile loss function. As this functional form is an asymmetric L_1 loss function proportional to the negative log density of the asymmetric Laplace distribution ($\mathcal{ALD}$), the connection allows researcher to recast the quantile regression as a maximum likelihood problem of the linear model $Y_i = \mathbf{X}_i\beta_{(\tau)} + \epsilon_{i,(\tau)}$ where $\epsilon_{i,(\tau)} \sim \mathcal{ALD}(\tau, 0, \sigma_{(\tau)})$ ¹¹. The working likelihood is of the form

$$f(Y | \mathbf{X}, \beta_{(\tau)}, \sigma_{(\tau)}, \tau) = \frac{\tau^n(1-\tau)^n}{\sigma_{(\tau)}^n} \exp \left\{ - \sum_{i=1}^n \frac{\rho_\tau(Y_i - \mathbf{X}_i\beta)}{\sigma_{(\tau)}} \right\}. \quad (\text{D.3})$$

The asymmetric Laplace distribution is known to be expressible as a scale mixture of normals (Kotz et al., 2012), we thus can rewrite $\epsilon_{i,(\tau)}$ as follows

$$\epsilon_{i,(\tau)} = \theta_{(\tau)} z_{i,(\tau)} + \kappa_{(\tau)} \sqrt{\sigma_{(\tau)} z_{i,(\tau)}} u_i, \quad \text{with } \theta_{(\tau)} = \frac{1-2\tau}{\tau(1-\tau)} \text{ and } \kappa_{(\tau)}^2 = \frac{2}{\tau(1-\tau)},$$

where $z_{i,(\tau)} = \sigma_{(\tau)} \zeta_{i,(\tau)}$ with $\zeta_{i,(\tau)} \sim \text{Exp}(1)$, and $u_i \sim N(0, 1)$.

As a result, the Bayesian quantile regression model has the following representation

$$Y_i = \mathbf{X}_i\beta_{(\tau)} + \theta_{(\tau)} z_{i,(\tau)} + \kappa_{(\tau)} \sqrt{\sigma_{(\tau)} z_{i,(\tau)}} u_i, \quad \text{for } i = 1, \dots, n. \quad (\text{D.4})$$

This leads to the following likelihood function:

$$f(Y | \mathbf{X}, \beta_{(\tau)}, \sigma_{(\tau)}, \mathbf{z}_{(\tau)}, \tau) \propto \exp \left\{ - \sum_{i=1}^n \frac{(Y_i - \mathbf{X}_i\beta_{(\tau)} - \theta_{(\tau)} z_{i,(\tau)})^2}{2\kappa_{(\tau)}^2 \sigma_{(\tau)} z_{i,(\tau)}} \right\} \prod_{i=1}^n \frac{1}{\sqrt{\sigma_{(\tau)} z_{i,(\tau)}}}. \quad (\text{D.5})$$

¹¹ $\epsilon_{i,(\tau)}$ follows asymmetric Laplace distribution with density

$$f_{\mathcal{ALD}}(\epsilon_{(\tau)}) = \frac{\tau(1-\tau)}{\sigma_{(\tau)}} \exp \{ -\rho_\tau(\epsilon_{(\tau)}) / \sigma_{(\tau)} \}.$$

We assume the priors as below (Kozumi and Kobayashi, 2011)

$$\beta_{(\tau)} \sim N\left(0, \mathbf{\Sigma}_{0,(\tau)}\right), \quad (\text{D.6})$$

$$z_{i,(\tau)} \sim \text{Exp}\left(\sigma_{(\tau)}\right) \propto \sigma_{(\tau)}^{-1} \exp\left\{-\sigma_{(\tau)}^{-1} z_{i,(\tau)}\right\} \forall i = 1, \dots, n, \quad (\text{D.7})$$

$$\sigma_{(\tau)} \sim \text{Inv-Gamma}\left(r_{0,(\tau)}, s_{0,(\tau)}\right) \propto \left(\sigma_{(\tau)}^{-1}\right)^{r_{0,(\tau)}+1} \exp\left\{-s_{0,(\tau)} \sigma_{(\tau)}^{-1}\right\}, \quad (\text{D.8})$$

where for simplicity, $\mathbf{\Sigma}_{0,(\tau)} = \mathbf{D}_{\lambda,(\tau)} = \lambda \times \mathbf{I}_p$ where λ is fixed and known for all τ .

The conditional posteriors are of the form

$$\beta_{(\tau)} \mid \bullet \sim N_p\left(\mu_{\beta,(\tau)}, \mathbf{\Sigma}_{\beta,(\tau)}\right), \quad (\text{D.9})$$

$$z_{i,(\tau)} \mid \bullet \sim \text{GIG}\left(\frac{1}{2}, a_{i,(\tau)}, b_{i,(\tau)}\right) \propto z_{i,(\tau)}^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}\left(a_{i,(\tau)} z_{i,(\tau)} + b_{i,(\tau)} z_{i,(\tau)}^{-1}\right)\right\}, \quad \forall i = 1, \dots, n, \quad (\text{D.10})$$

$$\sigma_{(\tau)} \mid \bullet \sim \text{Inv-Gamma}\left(r_{\sigma,(\tau)}, s_{\sigma,(\tau)}\right) \propto \left(\sigma_{(\tau)}^{-1}\right)^{r_{\sigma,(\tau)}+1} \exp\left\{-s_{\sigma,(\tau)} \sigma_{(\tau)}^{-1}\right\}, \quad (\text{D.11})$$

where

$$\mathbf{\Sigma}_{\beta,(\tau)} = \left(\mathbf{X}^\top \mathbf{U}^{-1} \mathbf{X} + \mathbf{\Sigma}_{0,(\tau)}^{-1}\right)^{-1} \text{ and } \mu_{\beta,(\tau)} = \mathbf{\Sigma}_{\beta,(\tau)} \times \mathbf{X}^\top \mathbf{U}^{-1} \left(\mathbf{Y} - \theta_{(\tau)} \mathbf{z}_{(\tau)}\right),$$

$$\mathbf{U} = \left(\sigma_{(\tau)} \kappa_{(\tau)}^2\right) \times \text{diag}\left(\mathbf{z}_{(\tau)}\right), \quad \mathbf{z}_{(\tau)} = \left(z_{1,(\tau)}, \dots, z_{n,(\tau)}\right)'$$

$$a_{i,(\tau)} = \frac{1}{\sigma_{(\tau)}} \left(2 + \frac{\theta_{(\tau)}^2}{\kappa_{(\tau)}^2}\right) \text{ and } b_{i,(\tau)} = \frac{\left(Y_i - \mathbf{X}_i \beta_{(\tau)}\right)^2}{\sigma_{(\tau)} \kappa_{(\tau)}^2},$$

$$r_{\sigma,(\tau)} = r_{0,(\tau)} + \frac{3n}{2} \text{ and } s_{\sigma,(\tau)} = s_{0,(\tau)} + \sum_{i=1}^n \frac{\left(Y_i - \mathbf{X}_i \beta_{(\tau)} - \theta_{(\tau)} z_{i,(\tau)}\right)^2}{2\kappa_{(\tau)}^2 z_{i,(\tau)}} + \sum_{i=1}^n z_{i,(\tau)}.$$

D.2 Bayesian Quantile Regression with the Adaptive Lasso

Bayesian Quantile Regression with the Adaptive Lasso is a Bayesian hierarchical model given by

$$Y_i = \mathbf{X}_i \beta_{(\tau)} + \theta_{(\tau)} z_{i,(\tau)} + \kappa_{(\tau)} \sqrt{\sigma_{(\tau)} z_{i,(\tau)}} u_i, \quad \text{for } i = 1, \dots, n, \quad (\text{D.12})$$

$$u_i \sim N(0, 1), \quad (\text{D.13})$$

$$z_{i,(\tau)} \sim \text{Exp}(\sigma_{(\tau)}) \propto \sigma_{(\tau)}^{-1} \exp\left\{-\sigma_{(\tau)}^{-1} z_{i,(\tau)}\right\} \forall i = 1, \dots, n \quad (\text{D.14})$$

$$\beta_{j,(\tau)}, v_{j,(\tau)} \mid \sigma_{(\tau)}, \lambda_{j,(\tau)}^2 \sim \frac{1}{\sqrt{2\pi v_{j,(\tau)}}} \exp\left\{-\frac{\beta_{j,(\tau)}^2}{2v_{j,(\tau)}}\right\} \frac{\sigma_{(\tau)}^{-1}}{2\lambda_{j,(\tau)}^2} \exp\left\{\frac{-\sigma_{(\tau)}^{-1}}{2\lambda_{j,(\tau)}^2} v_{j,(\tau)}\right\}, \quad (\text{D.15})$$

$$\lambda_{j,(\tau)}^2 \sim \text{Inv-Gamma}\left(c_{0,(\tau)}, d_{0,(\tau)}\right) \propto \left(\frac{1}{\lambda_{j,(\tau)}^2}\right)^{c_{0,(\tau)}+1} \exp\left\{-\frac{d_{0,(\tau)}}{\lambda_{j,(\tau)}^2}\right\}, \quad (\text{D.16})$$

$$\sigma_{(\tau)} \sim \text{Inv-Gamma}\left(r_{0,(\tau)}, s_{0,(\tau)}\right) \propto \left(\sigma_{(\tau)}^{-1}\right)^{r_{0,(\tau)}+1} \exp\left\{-s_{0,(\tau)} \sigma_{(\tau)}^{-1}\right\} \quad (\text{D.17})$$

The conditional posteriors ([Alhamzawi et al., 2012](#)) are of the form

$$z_{i,(\tau)} \mid \bullet \sim \text{GIG}\left(\frac{1}{2}, a_{i,(\tau)}, b_{i,(\tau)}\right) \propto z_{i,(\tau)}^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}\left(a_{i,(\tau)} z_{i,(\tau)} + b_{i,(\tau)} z_{i,(\tau)}^{-1}\right)\right\}, \quad \forall i = 1, \dots, n, \quad (\text{D.18})$$

$$\beta_{j,(\tau)} \mid \bullet \sim N\left(\mu_{\beta_j,(\tau)}, \Sigma_{\beta_j,(\tau)}\right), \quad \forall j = 1, \dots, p, \quad (\text{D.19})$$

$$v_{j,(\tau)} \mid \bullet \sim \text{GIG}\left(\frac{1}{2}, \frac{\sigma_{(\tau)}^{-1}}{\lambda_{j,(\tau)}^2}, \beta_{j,(\tau)}^2\right) \propto v_{j,(\tau)}^{-\frac{1}{2}} \exp\left\{-\frac{1}{2}\left(\frac{\sigma_{(\tau)}^{-1}}{\lambda_{j,(\tau)}^2} v_{j,(\tau)} + \beta_{j,(\tau)}^2 v_{j,(\tau)}^{-1}\right)\right\}, \quad (\text{D.20})$$

$$\sigma_{(\tau)} \mid \bullet \sim \text{Inv-Gamma}\left(r_{\sigma,(\tau)}, s_{\sigma,(\tau)}\right) \propto \left(\sigma_{(\tau)}^{-1}\right)^{r_{\sigma,(\tau)}+1} \exp\left\{-s_{\sigma,(\tau)} \sigma_{(\tau)}^{-1}\right\}, \quad (\text{D.21})$$

$$\lambda_{j,(\tau)}^2 \mid \bullet \sim \text{Inv-Gamma}\left(c_{0,(\tau)} + 1, d_{0,(\tau)} + \sigma_{(\tau)}^{-1} v_{j,(\tau)} / 2\right), \quad (\text{D.22})$$

where

$$a_{i,(\tau)} = \frac{1}{\sigma_{(\tau)}} \left(2 + \frac{\theta_{(\tau)}^2}{\kappa_{(\tau)}^2}\right) \text{ and } b_{i,(\tau)} = \frac{(Y_i - \mathbf{X}_i \beta_{(\tau)})^2}{\sigma_{(\tau)} \kappa_{(\tau)}^2},$$

$$\Sigma_{\beta_j,(\tau)} = \left[\left(\sigma \kappa_{(\tau)}^2\right)^{-1} \sum_{i=1}^n x_{ij}^2 z_{i,(\tau)}^{-1} + v_{j,(\tau)}^{-1} \right]^{-1},$$

$$\mu_{\beta_j,(\tau)} = \Sigma_{\beta_j,(\tau)} \left(\sigma \kappa_{(\tau)}^2\right)^{-1} \sum_{i=1}^n \left(Y_i - \theta_{(\tau)} z_{i,(\tau)} - \sum_{k=1, k \neq j}^p x_{ik} \beta_{k,(\tau)} \right) x_{ij}^2 z_{i,(\tau)}^{-1},$$

$$r_{\sigma,(\tau)} = r_{0,(\tau)} + \frac{3n}{2} + p \text{ and } s_{\sigma,(\tau)} = s_{0,(\tau)} + \sum_{i=1}^n \frac{(Y_i - \mathbf{X}_i \beta_{(\tau)} - \theta_{(\tau)} z_{i,(\tau)})^2}{2\kappa_{(\tau)}^2 z_{i,(\tau)}} + \sum_{i=1}^n z_{i,(\tau)} + \sum_{j=1}^p \frac{v_{j,(\tau)}}{2\lambda_j^2}.$$

D.3 On Implementation

The **bayesQR** R package ([Benoit and Van den Poel, 2017](#)) provides the implementation of efficient Gibbs sampler for both Bayesian Quantile Regression and Bayesian Quantile Regression with the Adaptive Lasso outlined above. In addition, the core procedure is programmed in Fortran to speed

up the computational time. Therefore, this package can be utilised straightforward to estimate multiple conditional quantiles, which then be used to approximate the condistional distributions of potential outcomes in our proposed BADR framework.

Alternatively, Variational Inference algorithm could be used for Bayesian quantile regression with/without the regularisation ([Lim et al., 2020](#); [Guo, 2019](#)), which helps improving the speed of Gibbs sampling while maintaining a comparable accuracy in terms of MSE.

E Implementation of Other Estimators in Simulation Study

E.1 Existing Approaches

Firpo's Inverse Probability Weighted method (FIPW)

Firpo's Inverse Probability Weighted (FIPW) method (Firpo, 2007) involves a two-step estimator. First, the propensity score is estimated nonparametrically as a logistic power series whose degree increases with sample size. In the second step, the quantiles are estimated by minimising an inverse probability weighted check loss function. These weights reflect the fact that the distribution of the covariates differs in the two groups.

Algorithm 2: FIPW Approach to Estimate QTEs

Data: $\{Y_i, T_i, \mathbf{X}_i\}_{i=1}^n, \tau \in (0, 1)$

Result: $\widehat{QTE}(\tau)$

- 1 *Step(1).* Estimate propensity score $\hat{\pi}(x) = \text{expit}(H_K(x)' \hat{p}_K)$ where

$$\hat{p}_K := \underset{p \in \mathbb{R}^K}{\text{argmax}} \frac{1}{N} \sum_{i=1}^N \{T_i \cdot \log(\text{expit}(H_K(X_i)'p)) + (1 - T_i) \cdot \log(1 - \text{expit}(H_K(X_i)'p))\}$$

- 2 *Step(2).* Derive $\hat{q}_1(\tau)$ and $\hat{q}_0(\tau)$ as the solution to

$$\hat{q}_1(\tau) := \underset{q}{\text{argmin}} \sum_{i=1}^N \frac{T_i}{N \cdot \hat{\pi}(\mathbf{X}_i)} \rho_\tau(Y_i - q) \text{ and } \hat{q}_0(\tau) := \underset{q}{\text{argmin}} \sum_{i=1}^N \frac{1 - T_i}{N \cdot (1 - \hat{\pi}(\mathbf{X}_i))} \rho_\tau(Y_i - q)$$

where $\rho_\tau(a) = a \cdot (\tau - \mathbb{1}\{a \leq 0\})$ is the check function.

- 3 Calculate $\widehat{QTE}(\tau) = \hat{q}_1(\tau) - \hat{q}_0(\tau)$.
-

Targeted Maximum Likelihood Estimation method (TMLE)

The estimation procedure of Targeted Maximum Likelihood Estimation (TMLE) method (Díaz, 2017) includes three steps. First, the propensity score and the conditional distribution of the outcome are estimated; second, the quantiles are estimated based on the current cdf of the outcome; and third, the conditional distribution of the outcome is updated based on an exponential submodel for the density of the outcome. The last two steps are iterated until convergence.

Algorithm 3: TMLE Approach to Estimate QTEs

Data: $\{Y_i, T_i, \mathbf{X}_i\}_{i=1}^n, \tau \in (0, 1)$

Result: $\widehat{QTE}(\tau)$

- 1 *Step (1). Initialize:* Obtain initial estimates $\hat{\pi}$ and $\hat{G}$ of π_0 and G_0 .
- 2 *Step (2). Compute $\hat{q}_1(\tau)$:* For the current estimate $\hat{G}$, compute

$$\hat{F}(y) = \frac{1}{n} \sum_{i=1}^n \hat{G}(y \mid 1, X_i) \text{ and } \hat{q}_1(\tau) = \inf\{y : \hat{F}(y) \geq \tau\}$$

- 3 *Step (3). Update $\hat{G}$:* Let $\hat{g}$ denote the density associated to $\hat{G}$.
- 4 (a) Consider the exponential submodel:

$$\hat{g}_\epsilon(y \mid 0, x) = c(\epsilon, \hat{g}) \exp\{\epsilon \hat{H}_{\hat{\eta}, \hat{\theta}(z)}\} \hat{g}(y \mid 0, x)$$

where $c(\epsilon, \hat{g})$ is a normalizing constant and

$$\hat{H}_{\hat{\eta}, \hat{\theta}(z)} := \frac{1}{\hat{\pi}(X)} \{\mathbb{1}_{(-\infty, \hat{\theta}]}(y) - \hat{G}(\hat{\theta} \mid 0, x)\}$$

is the score of the model.

- 5 (b) Estimate ϵ

$$\hat{\epsilon} = \operatorname{argmax} \sum_{i=1}^n (1 - T_i) \log \hat{g}_\epsilon(Y_i \mid 0, X_i)$$

- 6 (c) Calculate $\hat{g}_\epsilon(Y_i \mid 0, X_i)$ as the updated estimator of g
 - 7 *Step (4). Iterate:* Let $\hat{g} = \hat{g}_\epsilon$ and iterate steps 2-3 until convergence.
 - 8 Derive $\hat{q}_0(\tau)$ similarly.
 - 9 Calculate $\widehat{QTE}(\tau) = \hat{q}_1(\tau) - \hat{q}_0(\tau)$.
-

Localized Debiased Machine Learning method (LDML)

The Localized Debiased Machine Learning (LDML) method (Kallus et al., 2024) is also motivated by the efficient estimation equation, but Inverse Probability Weighted (IPW) estimates are used as rough initial guessed values for $\hat{q}_1$ and $\hat{q}_0$. Then, these values are used to localize the estimation of conditional distributions $\hat{G}(y \mid 0, \mathbf{X})$ and $\hat{G}(y \mid 1, \mathbf{X})$, respectively. This approach aims to refine the IPW estimate while obviating the need to estimate a continuum of continuum nuisances. The main algorithm includes two parts: *three-way-cross-fold nuisance estimation* and *solving the estimating equation*.

Algorithm 4: LDML Approach to Estimate QTEs

Data: $\{Y_i, T_i, \mathbf{X}_i\}_{i=1}^n, \tau \in (0, 1)$

- 1 *Part(1)*. Three-way-cross-fold nuisance estimation
- 2 Fix integers $K \geq 3$ and $1 \leq K' \leq K - 2$.
- 3 Randomly permute the data indices and let $\mathcal{D}_k$ be a random even K -fold split of the data.
- 4 **for** $k = 1, \dots, K$ **do**
- 5 (a) Set $\mathcal{H}_{k,1} = \{1, \dots, K' + \mathbb{1}[k \leq K']\} \setminus \{k\}$ and $\mathcal{H}_{k,2} = \{K' + \mathbb{1}[k \leq K'] + 1, \dots, K\} \setminus \{k\}$.
- 6 (b) Use only the data in $\mathcal{D}_k^{C,1} = \{\mathbf{X}_i : i \in \bigcup_{k' \in \mathcal{H}_{k,1}}\}$ to construct $\hat{q}_{1,init}^{(k)}$.
- 7 (c) Use only the data in $\mathcal{D}_k^{C,2} = \{\mathbf{X}_i : i \in \bigcup_{k' \in \mathcal{H}_{k,2}}\}$ to construct $\hat{G}_1^{(k)}(\cdot, \hat{q}_{1,init}^{(k)})$.
- 8 (d) Use $\mathcal{D}_k^{C,1} \cap \mathcal{D}_k^{C,2}$ to construct estimator $\hat{\pi}^{(k)}$.
- end**
- 9 *Part(2)*. Solving the average of the estimate in each fold to obtain $\hat{q}_1(\tau)$

$$\frac{1}{N} \sum_{k=1}^K \sum_{i \in \mathcal{D}_k} \psi(\mathbf{X}_i; q, \hat{G}_1^{(k)}(\mathbf{X}_i, \hat{q}_{1,init}^{(k)}), \hat{\pi}^{(k)}(\mathbf{X}_i)) = 0$$

- 10 Derive $\hat{q}_0(\tau)$ using two-part procedure similarly.
 - 11 Calculate $\widehat{QTE}(\tau) = \hat{q}_1(\tau) - \hat{q}_0(\tau)$.
-

Bayesian nonparametric method (BNP)

Bayesian nonparametric (BNP) method (Xu et al., 2018) is a fully Bayesian nonparametric (BNP) approach to estimate QTEs. The estimation procedure includes three steps. First, the propensity score is estimated using a logit BART model. Then, the conditional distribution of the potential outcome given a BART posterior sample of the PS in each treatment group is modelled separately using a Dirichlet process mixture (DPM) of multivariate normals model. Finally, marginalizing the estimated conditional distribution over the population distribution of the confounders using Bayesian bootstrap (Rubin, 1981). Details of implementation using **BNPqte** R package can be found in Luo and Daniels (2021).

Algorithm 5: BNP Approach to Estimate QTEs

Data: $\{Y_i, T_i, \mathbf{X}_i\}_{i=1}^n, \tau \in (0, 1)$

- 1 Fit a binary BART model on $\{T_i, X_i\}_{i=1}^n$ and obtain B posterior samples $\{H^{-1}(\pi_i^{\{b\}})\}_{i,b=1}^{n,B}$.
- 2 Create a set of grid points of Y values: $(q_1, \dots, q_S)$.
- 3 **for** $b = 1, \dots, B$ **do**
- 4 **for** $t = 0, 1$ **do**
- 5 Fit a DPM of bivariate normals on $\{Y_i, H^{-1}(\pi_i^{\{b\}})\}_{i:T_i=t}$.
- 6 Use Blocked Gibbs sampler to obtain L posterior samples.
- 7 Calculate $\{F^{\{bl\}}(q_s \mid H^{-1}(\pi_i^b), T = t)\}_{i,s,l=1}^{n,S,L}$.
- 8 **end**
- 9 Sample $(u_1^b, \dots, u_n^b)$ from $\mathcal{Dir}(1, \dots, 1)$
- 10 **for** $l = 1, \dots, L$ **do**
- 11 Calculate the CDF of $Y^{(t)}$ as follows:
$$SF_t^{bl}(q_s) = \sum_{i=1}^n u_i^b F^{kl}(q_s \mid H^{-1}(\pi_i^b), T = t), \text{ where } 1 \leq s \leq S \text{ and } t \in \{0, 1\}$$
- 12 Find a grid point $q_t^{bl}(\tau)$ such that $F_t^{kl}(q_t^{bl}(\tau)) = \tau$ for $t \in \{0, 1\}$.
- 13 The τ^{th} quantile from the CDF $F_t^{bl}(\cdot)$ is $q_{t,\tau}^{bl}$ for the group $T = t$.
- 14 **end**
- 15 **end**
- 16 Derive $\hat{q}_1(\tau)$ and $\hat{q}_0(\tau)$ as follows

$$\hat{q}_1(\tau) = \frac{1}{BL} \sum_b^B \sum_{l=1}^L q_1^{bl}(\tau) \quad \text{and} \quad \hat{q}_0(\tau) = \frac{1}{BL} \sum_b^B \sum_{l=1}^L q_0^{bl}(\tau)$$

- 14 Calculate $\widehat{QTE}(\tau) = \hat{q}_1(\tau) - \hat{q}_0(\tau)$.
-

E.2 Variants of the Proposed Approach

Bayesian Outcome Modelling (BOM)

Bayesian Outcome Modelling is an outcome-regression-based approach that omits the treatment assignment model. Instead, it solely focuses on estimating the conditional distribution through multiple Bayesian quantile regressions in the outcome model of each treatment group. Shrinkage priors, akin to the doubly-robust approach, can be readily incorporated.

Algorithm 6: Bayesian Outcome Modelling to estimate QTE

Data: $\{Y_i, T_i, \mathbf{X}_i\}_{i=1}^n, \tau \in (0, 1)$ **Result:** $\widehat{QTE}^{om}(\tau)$

- 1 **for** $t = 0, 1$ **do**
- 2 Fit outcome model on $\{Y_i, \mathbf{X}_i\}_{i:T_i=t}$ and obtain B posterior samples $\{G^{(b)}(y | t, \mathbf{X})\}_{b=1}^B$.
- 3 Derive posterior mean $\hat{G}(y | t, \mathbf{X}) = \frac{1}{B} \sum_{b=1}^B G^{(b)}(y | t, \mathbf{X})$.
- end**
- 4 Derive $\hat{q}_1^{om}(\tau)$ and $\hat{q}_0^{om}(\tau)$ as the solutions to

$$\frac{1}{n} \sum_{i=1}^n \hat{G}(q_1 | 1, \mathbf{X}) = \tau \text{ and } \frac{1}{n} \sum_{i=1}^n \hat{G}(q_0 | 0, \mathbf{X}) = \tau$$

- 5 Calculate $\hat{\Delta}_\tau \equiv \widehat{QTE}^{om}(\tau) = \hat{q}_1^{om}(\tau) - \hat{q}_0^{om}(\tau)$.
-

Bayesian Propensity Score Analysis (BPSA)

Bayesian Propensity Score Analysis is a treatment-assignment-based approach, which involves fitting the treatment assignment and then employs multiple Bayesian quantile regressions to model the conditional distribution of the outcome given the posterior mean of the propensity score in each treatment group.

Algorithm 7: Bayesian Outcome Modelling to estimate QTE

Data: $\{Y_i, T_i, \mathbf{X}_i\}_{i=1}^n, \tau \in (0, 1)$ **Result:** $\widehat{QTE}^{ps}(\tau)$

- 1 Fit treatment assignment model on $\{T_i, \mathbf{X}_i\}_{i=1}^n$ and obtain B posterior samples $\{\pi^{(b)}(\mathbf{X})\}_{b=1}^B$.
- 2 Derive posterior mean from B posterior samples $\hat{\pi}(\mathbf{X}) = \frac{1}{B} \sum_{b=1}^B \pi^{(b)}(\mathbf{X})$.
- 3 **for** $t = 0, 1$ **do**
- 4 Fit outcome model on $\{Y_i, \hat{\pi}(\mathbf{X}_i)\}_{i:T_i=t}$ and obtain B posterior samples $\{G^{(b)}(y | t, \hat{\pi}(\mathbf{X}))\}_{b=1}^B$.
- 5 Derive posterior mean $\hat{G}(y | t, \mathbf{X}) = \frac{1}{B} \sum_{b=1}^B G^{(b)}(y | t, \hat{\pi}(\mathbf{X}))$.
- end**
- 6 Derive $\hat{q}_1^{ps}(\tau)$ and $\hat{q}_0^{ps}(\tau)$ as the solutions to

$$\frac{1}{n} \sum_{i=1}^n \hat{G}(q_1 | 1, \mathbf{X}) = \tau \text{ and } \frac{1}{n} \sum_{i=1}^n \hat{G}(q_0 | 0, \mathbf{X}) = \tau$$

- 7 Calculate $\hat{\Delta}_\tau \equiv \widehat{QTE}^{ps}(\tau) = \hat{q}_1^{ps}(\tau) - \hat{q}_0^{ps}(\tau)$.
-

F Additional Graphs in Empirical Illustration

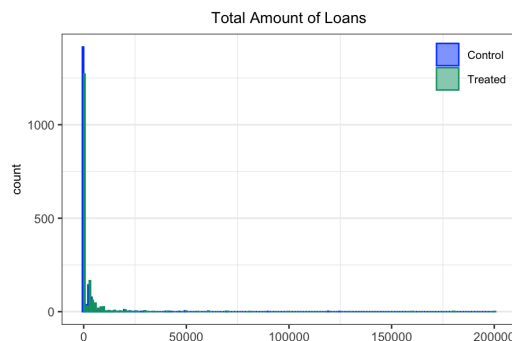


Figure F.1: Histogram of total amount of loans at household level in treated villages and control villages.

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